# Treasury Bill Market Dynamics: Strategic Analysis of System Structure, Actor Behavior, and Monitoring Framework

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*With AI assistance*

***Abstract*—This document provides a comprehensive analysis of the U.S. Treasury bill (T-bill) market as a critical control surface of the global dollar funding system. We examine the system structure, strategic behaviors of key market participants, and develop frameworks for monitoring systemic liquidity and confidence. The analysis maps T-bill markets as networks of feedback loops among diverse actors including the U.S. Treasury, Federal Reserve, domestic institutions, foreign central banks, and potential adversarial actors. We identify key latent state variables such as dollar liquidity stress, confidence in fiscal rollover, and market-making capacity, and propose observable indicators to monitor these hidden states. The document presents equilibrium models demonstrating how T-bill markets can exhibit multiple stable states and nonlinear transitions between regimes. Finally, we outline a practical Python-based monitoring toolkit and AI-assisted analysis framework for real-time surveillance of T-bill market health and early warning of systemic stress.**

***Index Terms*—Treasury bills, money markets, financial stability, systemic risk, liquidity monitoring, dollar funding, repo markets, equilibrium models, strategic actors, financial surveillance**

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## I. General Overview: System Structure & Stakes

### A. Mapping the T-Bill System

**Money markets, repo, and collateral chains:** T-bills (Treasury bills) are short-term U.S. government securities that play a pivotal role in dollar money markets. They serve as near-cash assets and top-quality collateral in repurchase agreements (repo) and other secured funding markets. Because of their liquidity and credit quality, T-bills are routinely re-used (rehypothecated) in collateral chains, "lubricating" the global financial system by allowing the same asset to back multiple loans [1]. A plentiful supply of T-bills supports high collateral *velocity*, whereas shortages or hoarding of T-bills can constrict this circulation. In essence, T-bills function as the *safe near-money* at the core of dollar funding markets, providing the benchmark yield for very short maturities and a secure store of cash for investors.

**Dollar liquidity and FX transmission:** The T-bill market is tightly linked to global dollar liquidity conditions. Foreign institutions often hold T-bills as part of their USD liquidity buffers or for FX reserve management. In offshore dollar funding markets (e.g. Eurodollar markets), T-bills frequently serve as swap or repo collateral to obtain USD funding. Thus, stresses in dollar funding (say, a sudden scramble for cash) are reflected in T-bill pricing. For example, during the March 2020 "dash-for-cash" turmoil, investors worldwide sold even safe assets to raise dollars [2], causing unusual yield spikes in Treasuries until central bank liquidity backstops were deployed. Conversely, abundant dollar liquidity (such as after Fed interventions) tends to drive T-bill yields down and can ease USD funding strains globally. Movements in T-bill yields also transmit into FX markets: a rise in T-bill rates relative to other countries' rates can attract global capital (strengthening the USD), whereas an anomalous divergence—such as rising T-bill yields *and* a weakening USD—may signal foreign holders reducing exposure (selling Treasuries and USD simultaneously). This makes the T-bill market a sensitive barometer of cross-border dollar flow dynamics.

**Treasury refinancing and fiscal operations:** The U.S. Treasury relies on T-bills as a key tool for managing short-term funding needs and cash flow. Bills (maturities up to 1 year) are issued frequently and in flexible amounts, which allows the Treasury to respond to seasonal cash imbalances or unexpected financing needs more readily than with longer-term bonds. In the aggregate, T-bills constitute a significant portion of the government's overall debt issuance and must be continually rolled over at auction. This rollover creates a recurrent test of market appetite. A well-functioning T-bill market, with stable demand at auctions, ensures the Treasury's ability to refinance maturing obligations at low cost, preserving fiscal capacity. T-bills are also integral to the Federal Reserve's operations: for instance, the Fed holds and purchases bills for implementing monetary policy (such as quantitative easing or, in 2019, bill purchases to rebuild bank reserves). Additionally, the Treasury's cash management bills (very short maturities) are used to bridge funding gaps around tax dates or debt-ceiling constraints. All these functions mean that T-bill market conditions feed directly into the government's financial position and its flexibility in managing public finances.

**Bills vs. notes/bonds; domestic vs. foreign; official vs. private holders:** Treasury bills differ from longer-term notes and bonds in duration and investor base. Bills are zero-coupon instruments that mature in one year or less, making them much less sensitive to interest-rate fluctuations than notes/bonds. Bills are preferred by money-market investors (such as money market funds, short-term investment pools, corporations managing cash) who value safety and liquidity over yield. In contrast, notes and bonds (with maturities up to 30 years) appeal to investors seeking term yield or specific portfolio matching. Foreign holders of Treasuries tend to include both "official" holders (central banks and sovereign wealth funds) and private investors. Official foreign institutions often hold significant bill positions as part of foreign exchange reserves for liquidity needs, while private foreign investors might seek longer Treasuries for yield or currency speculation. Notably, in periods of stress, these patterns can shift: foreign central banks may liquidate longer-term Treasuries but park more in bills for safety, whereas domestic institutions (like U.S. money market funds and banks) might increase bill holdings when risk aversion rises [3] [4]. For example, in Q1 2020, U.S. money market funds *increased* their holdings of T-bills by about $195 billion, even as foreigners were net sellers of longer-term Treasuries [3]. Such distinctions matter because the stability of the T-bill market can depend on which holders are dominant: official institutions may behave with different motives (e.g. stabilizing markets or managing exchange rates) compared to profit-driven private investors. Domestic holders might be influenced by regulatory/liquidity constraints, whereas foreign holders may respond to geopolitical or macroeconomic considerations.

**Disproportionate importance of bills (despite short duration):** Although T-bills have short maturities, they punch above their weight in systemic importance. First, the sheer volume of rollover means that any loss of confidence can immediately translate into refinancing strain for the Treasury. Bills typically account for a large share of Treasury's frequent issuance; if an auction goes poorly or if bid demand dwindles, it could force the Treasury to pay higher rates or, in an extreme case, raise questions about near-term funding—a situation unthinkable for U.S. Treasuries but nevertheless priced as tail-risk during debt ceiling standoffs [5]. Second, T-bills set a baseline risk-free rate for the economy's short end. Many other rates—from corporate commercial paper to bank deposit rates—take cues from prevailing T-bill yields. If bill yields become volatile or elevated due to market dislocations, it can transmit stress to all corners of short-term finance. Third, because bills are heavily used as collateral, their smooth functioning underpins repo markets; a scarcity of bills can drive unusual premiums (lower yields) as investors pay up for safe collateral [1] [6], whereas an glut or fear of bills (for example, if default risk is feared) can cause their yields to spike and destabilize repo rates. Finally, T-bills often serve as a safe haven: in risk-off episodes, investors flock to bills as the most liquid USD asset, which can stabilise broader markets by absorbing cash; but if even the bill market is strained, it indicates severe system-wide stress. In short, T-bills, though short-lived, act as the "tip of the spear" for the U.S. financial system's health—providing early warning signals and a primary interface for liquidity provision.

### B. Failure Modes & Stress Channels

Even in a highly creditworthy market like U.S. Treasuries, various failure modes can occur, generally categorized as *liquidity stress* vs. *solvency (credit) stress*, and as *market-functioning failure* vs. *macroeconomic regime change*. These are distinct but interrelated stress channels through which T-bill markets could destabilize the broader system.

**Liquidity stress vs. solvency stress:** A liquidity stress in the T-bill market means a sudden disruption in the ability to trade bills seamlessly at stable prices, usually due to a spike in the demand for cash (or shortage of cash) rather than doubts about the Treasury's credit. In a liquidity crunch, investors may engage in a fire-sale of even high-quality assets ("dash for cash"), causing yields to jump sharply and bid-ask spreads to widen. A historical example is March 2020: as the COVID-19 crisis intensified, investors sold virtually everything including T-bills for cash, leading to momentary spikes in T-bill yields and severe market illiquidity [2]. The Federal Reserve had to step in with massive purchases and liquidity facilities to restore order. Liquidity stress tends to be acute but reversible with central bank support. By contrast, solvency stress refers to market concerns about the borrower's ability or willingness to repay. In the context of U.S. Treasuries, this usually surfaces during debt ceiling impasses or fear of technical default. For instance, during the 2011 and 2013 debt-ceiling crises, T-bills maturing around the projected "X-date" (when the Treasury would run out of cash) traded at unusually elevated yields, reflecting a perceived default or delayed-payment risk [5]. In 2013, the default premium on one-month bills peaked at around 46 basis points (0.46%) above comparable-risk-free rates [5], an extraordinary deviation for an instrument normally viewed as virtually risk-free. Solvency stress in Treasuries is a more existential threat—if investors truly doubt the credit of the U.S. government, the entire edifice of the global financial system (which relies on Treasuries as the ultimate safe asset) is at risk. However, such episodes have so far been rare and quickly resolved politically. Notably, liquidity and solvency stresses can interact: a perceived credit risk (solvency issue) can freeze liquidity as counterparties refuse to trade, and conversely, extreme illiquidity can exaggerate price moves and give false signals of credit risk.

**Market-functioning failure vs. macroeconomic regime change:** A market-functioning failure is when the microstructure of the T-bill market breaks down. This includes events like an auction failure, a situation where there aren't enough bids to cover the amount offered, or severe dislocations where normally small spreads (like when-issued vs. auction yield, or repo rate vs. bill yield) blow out. For example, a "failed auction" is exceedingly rare for U.S. Treasuries, but a weak auction (low bid-to-cover, significant tail in yield) is a warning sign of market strain [7] [5]. Similarly, surging Treasury repo rates (well above policy rates) and frequent settlement fails are symptoms of market-functioning issues—often indicating that dealers' balance sheets are overburdened or that key intermediaries have stepped back. The spike in repo rates to nearly

10% intraday in September 2019 [6] is an example: it reflected a sudden scarcity of cash relative to Treasuries available (a micro market mismatch), which, absent Fed intervention, could have impaired short-term funding markets. Market-functioning failures are usually addressed by technical means: central bank operations, regulatory adjustments (e.g. relaxing leverage ratios temporarily), or Treasury debt management tweaks (such as increasing bill supply when there's collateral shortage).

By contrast, a macroeconomic regime change refers to a fundamental shift in the backdrop that alters the entire equilibrium of the T-bill market. This could be a transition from low inflation to high inflation regime, a loss of confidence in the U.S. dollar as global reserve currency, or a shift from ample liquidity to scarce liquidity as a deliberate policy. In such a case, T-bill yields might rise not just temporarily but structurally, and investor behavior (and expectations) undergoes a lasting change. For instance, the period from 2021–2023 saw soaring inflation and rapid Fed rate hikes, which moved the U.S. out of the zero-rate, ample-reserves regime that prevailed after 2008. In a high-inflation regime, T-bills, while still safe in nominal terms, carry more inflation risk—investors may demand higher yields, and the central bank's ability to backstop markets is constrained by the need to fight inflation. This trade-off between the inflation-fighting mandate and financial stability was largely absent in the prior low-inflation regime. If investors perceive that fiscal dominance looms (i.e., the central bank might sacrifice price stability to keep government financing costs down), that could usher in a new equilibrium with higher yields across even short maturities, reflecting a regime change in inflation expectations and risk premia. Another example of regime change is if the post-2008 "ample liquidity" paradigm (with trillions in bank reserves, abundant cash) shifts to a regime of quantitative tightening and scarce reserves: T-bill yields and repo dynamics might behave very differently, perhaps with more frequent cash shortages and volatile spikes in funding rates.

**Historical analogs and stress patterns:** History provides analogs for these stress channels. The "dash for cash" in March 2020 is a quintessential liquidity-driven market failure: it resembled the 2008 post-Lehman selloff, though in 2020 the distress was more about selling *everything* for cash than about counterparty credit. During that episode, correlations went to 1, and even normally inverse relationships reversed (Treasury prices fell alongside stocks initially) [3]. The Treasury market became "dysfunctional" enough that bid-ask spreads ballooned and price discovery stalled, prompting the Fed to initiate unlimited QE for Treasuries. Another analog is the 2008 crisis "run on repo," where concerns about collateral (MBS then, not Treasuries) led to a freeze in repo markets—by contrast, in 2020 the run was on *all* assets including Treasuries, highlighting that even the safest collateral can face liquidity runs if everyone demands cash simultaneously. On the solvency side, repeated U.S. debt ceiling impasses (2011, 2013, 2021, 2023) have given glimpses of market fear of technical default. Each time, yields on the at-risk bills jumped, only to normalize after political resolution; these are tiny tremors of a scenario so far averted. Internationally, one can point to events like the UK's September 2022 gilt crisis (triggered by forced selling by pension funds) as a case where a traditionally safe government bond saw a vicious spiral of yield increases due to a particular investor base's liquidity stress. While not about U.S. T-bills, it underscores how a feedback loop of selling and illiquidity can push yields to levels completely disconnected from policy rates or fundamental value in a short time, requiring a central bank intervention as a circuit breaker. All these analogs emphasize that the T-bill market's stability hinges on both the macro environment (are investors confident in the overall trajectory of policy and inflation?) and the microstructure (are dealers and funds able to intermediate flows smoothly?). Failures can manifest as sudden jumps in yields, dislocations in spreads, or evaporation of market depth; detecting those early and understanding their cause (liquidity vs. solvency, technical vs. fundamental) is crucial for an appropriate response.

Deliverable – System Diagram (conceptual): In summary, one can conceptualize the T-bill system as a network of actors and feedback loops. On one side, the U.S. Treasury is continuously feeding new T-bills into the market (to refinance debt and manage cash), and on the other side, a diverse set of investors absorb this supply: domestic money market funds, banks, corporates, households, foreign central banks, foreign private investors, dealers, etc. The Federal Reserve stands somewhat aside as a backstop/market-maker of last resort, ready to provide liquidity via repo or outright purchases in stress times.

Key feedback loops include: (1) Liquidity spiral loop: if stress causes selling and yields rise, dealers' balance sheet constraints may reduce their intermediation, further reducing liquidity—until a backstop (Fed or large liquidity provider) breaks the loop. (2) Safe-haven loop: in risk-off scenarios, more demand flows to T-bills (prices up, yields down), which can stabilize the system by satiating the need for safety; but if the shock is so large that even T-bills are sold (as in dash-for-cash), that loop fails and turns into a liquidity spiral instead. (3) Policy expectation loop: T-bill yields are closely tied to expectations of Fed policy. If investors expect Fed rate cuts, they pile into bills (pushing yields below the fed funds rate); if instead unexpected tightening is anticipated, bill yields jump. This loop means monetary policy signals quickly feed through to T-bill yields and thus to broader financial conditions. (4) Collateral loop: the availability of T-bills as repo collateral affects repo rates, which feed into leverage and funding for other asset markets; a shortage of bills (or sudden hoarding) can drive down repo rates (or cause fails), which in turn might prompt Treasury or Fed to adjust operations (e.g., Treasury increasing bill issuance, or Fed adjusting its reverse repo facility to set a floor). All these actors and loops create a dynamic system where T-bills are the control surface: interventions (by Treasury via issuance mix, or Fed via liquidity facilities) on this surface propagate through money markets and global finance. A failure in any link (e.g., dealers hitting balance sheet limits, or foreign

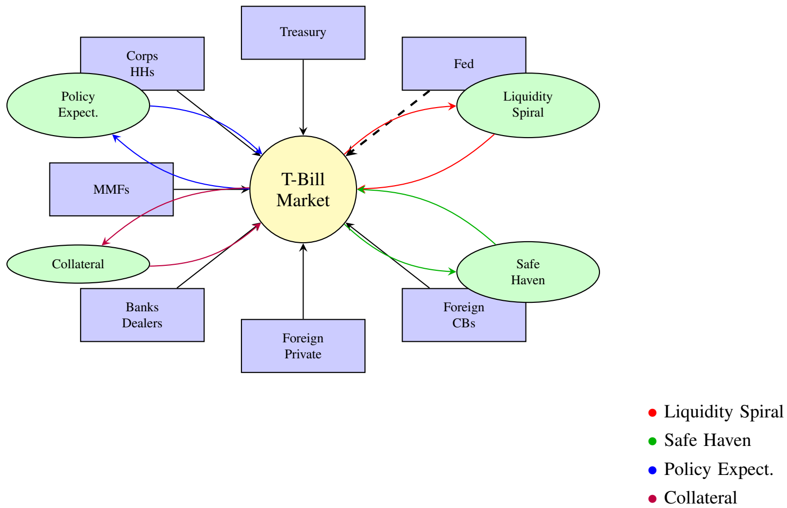


Fig. 1. *T-Bill System Network: Actors and Feedback Loops. The T-bill market sits at the center, with Treasury issuing supply and various investor classes providing demand. The Fed acts as backstop. Four key feedback loops (color-coded) create system dynamics.*

reserve managers suddenly selling bills) can propagate quickly, given the central role of T-bills in funding and collateral.

### PRIMER: U.S. GOVERNMENT DEBT INSTRUMENTS AND MARKET STRUCTURE

*Types of U.S. Government Securities*

**Regular Issuance Securities (Normal Government Operations):**

- **Treasury Bills (T-bills):** Short-term securities with maturities of 1 year or less (4, 8, 13, 17, 26, and 52 weeks). Sold at discount to face value, mature at par. No periodic interest payments. *Primary use:* Cash management, short-term funding, money market liquidity. *Key difference:* Zero-coupon structure makes them ideal for precise cash flow timing and repo collateral.
- **Treasury Notes (T-notes):** Medium-term securities with maturities of 2, 3, 5, 7, and 10 years. Pay semi-annual coupon interest at fixed rates. *Primary use:* Intermediate-term funding, benchmark for corporate bonds and mortgages. *Key difference:* Coupon payments provide steady income stream, making them attractive for insurance companies and pension funds.
- **Treasury Bonds (T-bonds):** Long-term securities with maturities of 20 and 30 years. Pay semi-annual coupon interest at fixed rates. *Primary use:* Long-term funding, duration matching for long-term liabilities. *Key difference:* Highest duration sensitivity, often used for asset-liability matching by life insurers.
- **TIPS (Treasury Inflation-Protected Securities):** Real return bonds with maturities of 5, 10, and 30 years. Principal adjusts with inflation (CPI), coupon rate applied to adjusted principal. *Primary use:* Inflation hedging for institutions and individuals. *Key difference:* Only Treasury security with built-in inflation protection.
- **I Bonds (Series I Savings Bonds):** Retail-only savings bonds with 30-year maturity and inflation adjustment. Limited to $10,000 per person annually. *Primary use:* Inflation protection for individuals, alternative to bank deposits. *Key difference:* Not tradeable, designed for buy-and-hold retail investors.

**Special Circumstances Securities:**

- **Cash Management Bills (CMBs):** Irregular, short-term bills (typically 7-50 days) issued to address temporary cash shortfalls or surpluses. *Circumstances:* Seasonal tax receipt patterns, unexpected government expenditures, bridging between regular auction cycles. *Example:* Large tax refund periods in February-April requiring additional short-term funding.
- **Floating Rate Notes (FRNs):** 2-year securities with quarterly rate resets tied to 13-week T-bill rates. *Circumstances:* Periods of high interest rate volatility when Treasury wants to reduce refinancing risk while attracting money market investors. *Issued 2014-2020:* During Federal Reserve policy normalization period.
- **STRIPS (Separate Trading of Registered Interest and Principal):** Zero-coupon securities created by "stripping" coupon and principal payments from regular Treasury bonds and notes. *Circumstances:* Market demand for zero-coupon instruments for precise duration matching. *Use case:* Pension funds matching specific liability dates.

- **State and Local Government Series (SLGS):** Special issues sold directly to state and local governments to help them comply with federal tax regulations on advance refundings. *Circumstances:* Municipal bond refunding transactions requiring temporary investment of proceeds.
- **Government Account Series (GAS):** Non-marketable securities held by federal government trust funds and accounts. *Circumstances:* Social Security Trust Fund, federal employee retirement funds investing surplus contributions. *Key feature:* Intragovernmental debt not traded in markets.
- **Debt Ceiling Crisis Instruments:** *Emergency measures when approaching debt limit:*
  - *Extraordinary measures:* Temporary suspension of investments in Civil Service Retirement Fund, Postal Service Retiree Health Benefits Fund
  - *Cash management:* Accelerated tax collection, delayed payments, use of Treasury General Account balances
  - *Technical default instruments:* In theory, could issue "premium bonds" above debt limit (never used)
- **Crisis/Wartime Special Issues:** Historically issued during major crises:
  - *War bonds:* World War I Liberty Bonds, World War II War Bonds with special patriotic features
  - *Savings bonds variants:* Special series during crises (e.g., post-9/11 Patriot Bonds)
  - *Emergency authority:* Treasury has broad powers under national emergency declarations to modify terms and issuance patterns

**Key Operational Differences:**

- **Liquidity:** T-bills most liquid (daily trading), bonds less liquid, SLGS/GAS non-tradeable
- **Price volatility:** Bills least volatile (short duration), bonds most volatile (long duration)
- **Investor base:** Bills (money funds, banks), Notes/Bonds (pension funds, foreign CBs), TIPS (real money investors)
- **Auction frequency:** Bills (weekly), Notes (monthly), Bonds (quarterly), Special issues (as needed)

*Understanding Coupon Payments*

**What are Coupon Payments?** Coupon payments are periodic interest payments made to bondholders during the life of a bond, separate from the final principal repayment at maturity. The term originates from historical bonds that had physical coupons that investors would clip and redeem for interest payments.

**How Coupon Payments Work:**

- **Coupon Rate:** The annual interest rate stated on the bond (e.g., 4.5% for a 10-year Treasury note)
- **Payment Calculation:** Semi-annual payment = (Coupon Rate × Face Value) ÷ 2
- **Example:** A $1,000 face value 10-year Treasury note with 4.5% coupon pays $22.50 every six months ($45 annually)
- **Payment Schedule:** U.S. Treasury securities pay coupons semi-annually on fixed dates
- **Final Payment:** At maturity, investors receive the final coupon payment plus the full principal ($1,000 face value)

**Coupon Rate vs. Yield:**

- **Coupon Rate:** Fixed percentage set at issuance, never changes
- **Current Yield:** Annual coupon payments ÷ current market price
- **Yield to Maturity (YTM):** Total return if held to maturity, including price appreciation/depreciation
- **Example:** A 4.5% coupon bond trading at $950 (below par) has current yield of 4.74% and YTM ¿ 4.74%

**Zero-Coupon vs. Coupon-Bearing Securities:**

- **Treasury Bills (Zero-Coupon):** No periodic payments; bought at discount, mature at full face value; all return comes from price appreciation
- **Treasury Notes/Bonds (Coupon-Bearing):** Regular semi-annual income stream plus principal at maturity; attracts income-focused investors
- **Cash Flow Timing:** Coupon bonds provide predictable cash flows throughout their life; T-bills provide single lump sum at maturity
- **Reinvestment Risk:** Coupon bonds face risk that coupon payments must be reinvested at potentially lower rates; T-bills avoid this risk

*Repurchase Agreements (Repo Markets)*

**What are Repurchase Agreements?** A repurchase agreement (repo) is a short-term borrowing mechanism where one party sells securities (typically Treasuries) to another party with a simultaneous agreement to repurchase the same securities at a slightly higher price on a specified future date. Economically, this functions as a collateralized loan where the securities serve as collateral.

**How Repo Transactions Work:**

- **Initial Sale:** Party A sells $100 million in T-bills to Party B for $100 million cash
- **Repurchase Agreement:** Party A agrees to buy back the same T-bills tomorrow for $100.01 million
- **Implicit Interest:** The $10,000 difference represents overnight interest (repo rate)
- **Collateral:** T-bills remain with Party B as security; if Party A defaults, Party B keeps the T-bills
- **Maturity:** Most repos are overnight, but can range from overnight to several months

**Key Participants and Their Motivations:**

- **Cash Lenders (Repo Buyers):**
  - *Money Market Funds:* Invest cash overnight in safe, short-term instruments
  - *Banks:* Manage excess deposits and regulatory liquidity requirements
  - *Corporations:* Park temporary cash surpluses securely
  - *Foreign Central Banks:* Invest dollar reserves in safe, liquid instruments
- **Cash Borrowers (Repo Sellers):**
  - *Primary Dealers:* Finance Treasury inventory, fund proprietary trading
  - *Hedge Funds:* Finance leveraged positions, implement arbitrage strategies
  - *Banks:* Meet short-term funding needs, manage balance sheet timing
  - *Securities Firms:* Finance market-making activities and client financing

**Types of Repo Transactions:**

- **General Collateral (GC) Repo:** Any acceptable Treasury security can be posted; repo rate close to overnight policy rate
- **Specific Collateral Repo:** Particular Treasury issue is specified; typically trades "special" (below GC rate) due to scarcity
- **Overnight Repo:** Single day maturity (most common)
- **Term Repo:** Multi-day or multi-week maturities
- **Open Repo:** Automatically rolls over each day until terminated
- **Tri-party Repo:** Uses custodian bank (BNY Mellon, JPMorgan) to manage collateral and cash transfers
- **Bilateral Repo:** Direct agreement between two parties

**Scale and Importance:**

- **Market Size:** Daily repo volumes of $4-5 trillion globally, with $3-4 trillion in U.S. Treasury repo
- **Critical Infrastructure:** Repo markets are essential plumbing for financial system liquidity
- **Fed Operations:** Federal Reserve uses repo operations to implement monetary policy (conducts $2+ trillion daily)
- **Benchmark Rates:** SOFR (Secured Overnight Financing Rate) based on Treasury repo transactions
- **Leverage Mechanism:** Enables financial institutions to leverage Treasury positions 10-50x through repo chains

**Why T-bills are Preferred Repo Collateral:**

- **No Accrued Interest:** Zero-coupon structure eliminates complex calculations
- **High Liquidity:** Can be easily sold if borrower defaults
- **Short Duration:** Minimal price volatility, stable collateral value
- **No Credit Risk:** Full faith and credit of U.S. government
- **Standardized:** Uniform terms and clearing mechanisms

**Repo Market Stress Indicators:**

- **Rate Spikes:** GC repo rates jumping above Fed policy rate (e.g., September 2019: repo rates hit 10%)
- **Fails to Deliver:** Inability to deliver promised securities (normal: ¡$100B daily, stress: ¿$500B)
- **Haircuts:** Increased collateral requirements (normal: 0-2% for Treasuries, stress: 5-10%+)
- **Term Structure Inversion:** Overnight rates above term rates (fear of rollover risk)
- **Bifurcated Markets:** Large spreads between dealer-to-dealer vs. dealer-to-customer rates

### *Auction Process and Schedule*

**Auction Schedule:**

- **T-bills:** Weekly auctions (4, 8, 13 weeks), monthly (17, 26 weeks), and bi-weekly (52 weeks)
- **T-notes:** Monthly auctions for most maturities, quarterly for some
- **T-bonds:** Quarterly auctions (February, May, August, November)
- **TIPS:** Quarterly or semi-annual depending on maturity

### *Market Participants and Auction Mechanics*

**Primary Dealers:** 24 large financial institutions with exclusive privileges and obligations:

- Required to bid competitively at Treasury auctions
- Provide liquidity in secondary markets
- Serve as counterparties to Federal Reserve operations
- Examples: JPMorgan, Goldman Sachs, Citigroup, foreign banks (Deutsche Bank, HSBC)

**Bidder Categories:**

- **Competitive bids:** Specify yield/price, may be rejected if too high
- **Non-competitive bids:** Accept whatever yield is determined (up to $10M for individuals)
- **Direct bidders:** Institutions bidding for themselves (banks, insurance companies, funds)
- **Indirect bidders:** Foreign central banks, sovereign wealth funds (via Federal Reserve)

**Auction Results Metrics:**

- **Bid-to-cover ratio:** Total bids divided by amount offered (healthy: $>$ 2.0 for notes/bonds)
- **Tail:** Difference between auction yield and pre-auction when-issued yield
- **Dealer take-down:** Percentage awarded to primary dealers vs. direct/indirect bidders

*Typical Uses and Functions*

- **Government funding:** Finance federal operations, refinance maturing debt
- **Liquidity management:** Short-term cash equivalent for institutions
- **Collateral:** Repo transactions, margin requirements, regulatory capital
- **Flight-to-safety:** Safe haven during market stress
- **Benchmark:** Risk-free rate for pricing other securities
- **Monetary policy transmission:** Fed policy affects Treasury yields across curve
- **Reserve management:** Foreign central banks hold as international reserves

*Key Market Statistics (2024)*

- **Outstanding debt:** Approximately $26 trillion total U.S. Treasury debt
- **T-bill outstanding:** Roughly $5-6 trillion (varies with Treasury cash management)
- **Daily trading volume:** $500+ billion across all Treasury securities
- **Foreign holdings:** Approximately $7.5 trillion (30% of total), with Japan ($1.1T) and China ($0.8T) as largest holders
- **Fed holdings:** $5+ trillion through QE programs and normal operations

This market structure creates the foundation for the strategic dynamics analyzed in the following sections. The concentration of auction participation among 24 primary dealers, the predictable auction calendar, and the massive scale of outstanding debt create both stability and potential vulnerabilities that actors can exploit or defend.

**Key Insights from Debt Distribution:**

- **Concentration in Medium-Term:** Treasury Notes (2-10 years) represent 55% of total debt, making this segment critical for funding
- **T-bills as Liquidity Hub:** 22% of debt in bills provides crucial short-term liquidity and repo collateral
- **Intragovernmental Holdings:** 28% of total debt held by government trust funds (primarily Social Security) represents captive demand
- **Maturity Concentration:** 2-year (16%) and 10-year (13%) notes are largest individual segments, making them key benchmarks
- **Rollover Risk:** Bills + 2-year notes (38% of marketable debt) must be refinanced within 2 years, creating significant rollover exposure
- **Foreign Vulnerability:** Foreign holdings concentrated in 2-10 year notes ($4-5T), making this segment sensitive to foreign central bank actions

**Trading Volume Insights:**

- **T-bills Dominate Trading:** Despite being only 22% of debt outstanding, T-bills account for 58% of annual trading volume ($30T vs $52T total), reflecting their role as the primary liquidity instrument
- **Turnover Ratios:** T-bills turn over 5.2x annually on average (some 4-week bills turn over 13x), while 30-year bonds turn over only 0.3x, showing the inverse relationship between maturity and liquidity
- **Policy Transmission:** 2-year notes have high turnover (2.0x) because they're the primary vehicle for trading Federal Reserve policy expectations
- **Benchmark Premium:** 10-year notes trade more actively (1.1x turnover) than their duration would suggest due to their role as the global benchmark
- **Repo Market Impact:** T-bill trading volume includes substantial repo transactions, making them the backbone of short-term funding markets
- **Crisis Amplification:** High turnover securities (bills, 2-year notes) can amplify volatility during stress as the same bonds change hands repeatedly

*Understanding Intergovernmental Debt (Government Account Series)*

The $6.8 trillion in Government Account Series (GAS) securities represents a unique category of U.S. debt that operates completely separately from marketable securities. This "debt held by the government itself" requires explanation because of its scale and strategic implications.

**What is Intergovernmental Debt?** Intergovernmental debt consists of special Treasury securities issued to federal government trust funds, accounts, and other government entities. These are non-marketable securities that cannot be traded in financial markets and exist purely as accounting entries between different parts of the federal government.

**Who Holds It and Why:**

- **Social Security Trust Funds ($2.8T):** Old-Age and Survivors Insurance (OASI) and Disability Insurance (DI) trust funds. When payroll taxes exceed current benefits, surpluses must be invested.
- **Federal Employee Retirement Funds ($1.1T):** Civil Service Retirement and Disability Fund, Thrift Savings Plan G Fund, Federal Employees Retirement System.
- **Military Retirement Fund ($0.7T):** Department of Defense Military Retirement Fund for future military pension obligations.
- **Medicare Trust Funds ($0.6T):** Hospital Insurance (Part A) and Supplementary Medical Insurance trust funds.
- **Highway Trust Fund ($0.1T):** Gasoline tax revenues invested until needed for highway projects.
- **Other Trust Funds ($1.5T):** Unemployment insurance, airport trust funds, nuclear waste disposal, black lung benefits, vaccine injury compensation, etc.

**How Intergovernmental Debt is Created:**

- **Legal Requirement:** Federal law requires trust fund surpluses to be invested in Treasury securities. Trust funds cannot hold cash or invest in private markets.

| Category | Maturity | Outstanding | % of Total | Annual Volume | Primary Uses/Investors |
|---|---|---|---|---|---|
| ***Treasury Bills (Bills)*** | | | | | |
| Bills | 4-week (1 month) | $1.1T | 4.2% | $14.3T | Ultra-short liquidity, Fed operations |
| Bills | 8-week (2 months) | $0.7T | 2.7% | $4.6T | Short-term cash management |
| Bills | 13-week (3 months) | $1.3T | 5.0% | $6.5T | Money market funds, benchmark rate |
| Bills | 17-week (4.5 months) | $0.5T | 1.9% | $1.4T | Intermediate cash management |
| Bills | 26-week (6 months) | $1.2T | 4.6% | $2.4T | Corporate treasuries, banks |
| Bills | 52-week (1 year) | $1.0T | 3.8% | $1.0T | Annual planning, foreign CBs |
| **Bills Subtotal** | | | **22.3%** | **30.2T** | **Short-term funding, liquidity** |
| ***Treasury Notes (Notes)*** | | | | | |
| Notes | 2-year | $4.2T | 16.2% | $8.4T | Short-term rates, Fed policy transmission |
| Notes | 3-year | $2.1T | 8.1% | $3.2T | Intermediate duration, bank portfolios |
| Notes | 5-year | $3.1T | 11.9% | $3.7T | Corporate benchmarks, pension ALM |
| Notes | 7-year | $1.6T | 6.2% | $1.4T | Insurance companies, duration gap |
| Notes | 10-year | $3.3T | 12.7% | $3.6T | Global benchmark, foreign reserves |
| **Notes Subtotal** | | | **55.0%** | **20.3T** | **Core funding, benchmarks** |
| ***Treasury Bonds (Bonds)*** | | | | | |
| Bonds | 20-year | $0.4T | 1.5% | $0.1T | Long duration, life insurers |
| Bonds | 30-year | $2.1T | 8.1% | $0.6T | Pension funds, long-term liabilities |
| **Bonds Subtotal** | | | **9.6%** | **0.7T** | **Long-term funding, duration** |
| ***Inflation-Protected Securities (TIPS)*** | | | | | |
| TIPS | 5-year | $0.4T | 1.5% | $0.1T | Short-term inflation hedge |
| TIPS | 10-year | $0.8T | 3.1% | $0.2T | Intermediate inflation protection |
| TIPS | 30-year | $0.5T | 1.9% | $0.1T | Long-term real return |
| **TIPS Subtotal** | | | **6.5%** | **0.4T** | **Inflation protection** |
| ***Other Marketable Securities*** | | | | | |
| FRNs | 2-year floating | $0.3T | 1.2% | $0.2T | Money markets, rate volatility hedge |
| **Other Subtotal** | | | **1.2%** | **0.2T** | **Specialized instruments** |
| ***Non-Marketable Securities*** | | | | | |
| GAS | Various | $6.8T | 26.2% | $0T | Social Security, federal trust funds |
| SLGS | Custom | $0.1T | 0.4% | $0T | State/local government compliance |
| I Bonds | 30-year | $0.1T | 0.4% | $0T | Retail inflation protection |
| Other | Various | $0.2T | 0.8% | $0T | Special purpose instruments |
| **Non-Marketable Subtotal** | | | **27.7%** | **0T** | **Intragovernmental, retail** |
| **TOTAL DEBT** | | | **100.0%** | **51.8T** | |
| **MARKETABLE DEBT** | | | **77.5%** | **51.8T** | |

TABLE I

*Distribution of Outstanding U.S. Government Debt by Instrument and Maturity. Figures represent approximate 2024 levels. Marketable debt (77.5%) trades in secondary markets, while non-marketable debt includes intragovernmental holdings and retail instruments.*

- **Special Issues:** Treasury creates non-marketable securities specifically for each trust fund, with terms designed to match the fund's needs.
- **Interest Rates:** GAS securities earn interest at rates calculated using a formula based on marketable Treasury securities, typically close to the average yield on outstanding Treasury debt.
- **Automatic Process:** When a trust fund has surplus cash (e.g., Social Security collecting more in payroll taxes than it pays in benefits), Treasury automatically issues GAS securities.
- **Book Entry Only:** These transactions are purely accounting entries – no physical securities exist, and no money changes hands between Treasury and the trust fund.

**How it's Paid Back:**

- **Redemption on Demand:** When a trust fund needs cash to pay benefits, Treasury redeems the GAS securities immediately at face value plus accrued interest.
- **General Fund Impact:** Treasury must fund these redemptions from general revenues, by issuing new marketable debt, or from current cash.
- **No Market Risk:** Unlike marketable securities, GAS securities have no price volatility – they always redeem at par value.
- **Automatic Rollover:** Maturing GAS securities are typically rolled over into new GAS securities unless the trust fund needs the cash for expenditures.

**Why Not Use Ordinary Bond Markets:**

- **Legal Restrictions:** Federal law specifically requires trust fund surpluses to be invested in Treasury securities, not private markets or other assets.
- **Fiduciary Responsibility:** Trust funds manage money for specific beneficiaries (retirees, disabled workers, etc.) and must avoid market risk that could jeopardize benefit payments.
- **Liquidity Requirements:** Trust funds need to convert investments to cash immediately when benefits are due. Marketable securities might trade at a loss.
- **Accounting Separation:** GAS securities maintain clear separation between trust fund assets and general govern-

ment operations for accounting and political purposes.

- **Interest Rate Stability:** GAS securities provide steady, predictable returns without the volatility of market-traded securities.
- **No Transaction Costs:** Book-entry system eliminates bid-ask spreads, dealer fees, and other trading costs that would reduce trust fund returns.

**Why Not Simply Use the Government as a "Bank"?** A logical question is why the government creates these internal debt instruments at all, rather than simply having a centralized treasury where all government money is pooled and managed. The answer involves fundamental design principles:

- **Fiduciary Separation:** Trust funds represent money that belongs to specific beneficiaries, not the general public. Debt instruments create a legal firewall between "their" money and general government operations.
- **Political Protection:** Makes it harder for Congress to spend trust fund money on other programs. Accessing trust fund money requires explicitly redeeming securities, creating a visible political act.
- **Actuarial Soundness:** Earning interest on surpluses maintains the self-funding principle of programs like Social Security. No return would make the system less actuarially sound.
- **Constitutional Compliance:** Treasury withdrawals require appropriation. Trust funds with dedicated securities demonstrate that specific appropriations are backed by specific assets.
- **Insurance Mathematics:** Trust funds operate like insurance systems requiring reserves that earn returns to remain solvent over decades.
- **Intergenerational Equity:** Today's surplus contributions should benefit those same workers when they retire, maintained through interest earnings.

**Strategic and Political Implications:**

- **Captive Financing:** Intragovernmental debt provides Treasury with \$6.8T in guaranteed financing that cannot "flee to safety" during market stress.
- **Future Pressure:** As trust funds (especially Social Security) shift from surplus to deficit, they will become net redeemers, forcing Treasury to find \$6.8T in replacement financing from markets.
- **Political Confusion:** Critics sometimes claim the government is "raiding" Social Security, but this is how the system is legally designed to work.
- **Debt Ceiling Impact:** Intergovernmental debt counts toward the debt ceiling, and "extraordinary measures" during debt ceiling crises often involve temporarily suspending GAS issuance.
- **Generational Transfer:** When trust funds redeem GAS securities, the burden shifts from future trust fund beneficiaries to future general taxpayers who must service the replacement marketable debt.

**Timeline Pressure:** The Social Security Trustees project that OASDI trust funds will begin running deficits around 2034, requiring net redemptions of GAS securities. This will force Treasury to replace up to \$2.8T of captive intergovernmental financing with marketable debt that must compete for private investment – potentially creating significant upward pressure on interest rates and Treasury funding costs.

## II. Strategic Actor Analysis

We now examine how different classes of rational actors might deliberately use or respond to the T-bill market as a "control surface" to achieve their objectives. We consider three archetypes: (1) Friendly or aligned actors, who have an interest in U.S. financial stability; (2) Neutral or self-interested actors, who care primarily about their own risk-adjusted returns; and (3) Hostile or adversarial actors, who might seek to weaken U.S. financial strength or send political signals, even at some cost to themselves. For each, we outline their potential objectives, the tools or strategies they could employ involving T-bills, the constraints or risks that limit their actions, and the expected system responses.

### A. Friendly/Aligned Actors

These include allies and partner countries (e.g. close reserve-currency holders), as well as large domestic institutions (major U.S. banks, primary dealers, or the Federal Reserve itself in a lender-of-last-resort capacity). Their overarching objective is to maintain stability and confidence in the U.S. Treasury market, as that underpins global financial stability. They prefer smooth functioning and are averse to causing disruptions.

Objectives: Ensure the continuous liquidity and reliability of T-bills; prevent fire-sale dynamics; support U.S. fiscal capacity by keeping borrowing costs reasonable; and generally uphold the dollar-based financial order. They may also want to signal confidence in the U.S. economy and discourage speculative attacks on the Treasury market.

Tools/Strategies: One key strategy is *portfolio smoothing* — i.e. adjusting their own purchases or sales of T-bills to counteract market volatility. For example, an aligned central bank might quietly buy more T-bills if yields are spiking (to add demand) or hold off on redemptions during a delicate period. Similarly, large U.S. banks under guidance could temporarily increase their holdings or use their balance sheet to intermediate (repo lending against Treasuries) when dealer liquidity is strained. *Maturity laddering* is another strategy: aligned investors structure their Treasury portfolios so that maturities are staggered, avoiding concentration of rollovers at a single moment. This reduces the risk of a refinancing cliff and makes their rollover behavior predictable, which stabilizes market expectations. *Coordinated reinvestment* is a softer form of intervention: for instance, a group of friendly central banks could collectively agree (often tacitly) to reinvest their maturing U.S. T-bills back into new T-bills during a stress period, rather than letting that money flee elsewhere. By doing so, they support demand at auctions without overtly announcing any "support program." In extreme cases, direct *Federal Reserve action* is possible (though the Fed is not an "actor" in the game-theoretic sense of market strategy, its

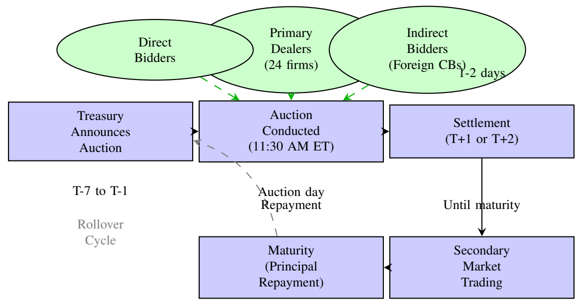


Fig. 2. *U.S. Treasury Auction and Lifecycle Process. Treasury securities follow a compact cycle from announcement through auction to secondary trading and maturity, with potential rollover.*

policy choices matter): the Fed can operate a standing repo facility or purchase bills ("quantitative easing") to put a floor under prices in a panic.

A critical aspect is *signaling vs. action*. Friendly actors often prefer to stabilize via credible signals rather than heavy-handed market operations, to avoid accusations of manipulation. For example, a major ally might publicly affirm confidence in U.S. Treasuries or state that they have no plans to significantly reduce their holdings—such verbal assurances can calm markets without any actual intervention. The U.S. Treasury and Federal Reserve also engage in signaling: statements that the Treasury has ample cash or that the Fed is "ready to use its tools" can by themselves deter panic. That said, if talk proves insufficient, these actors will follow through with action to maintain credibility.

Constraints: Friendly actors, while aligned with stability goals, face trade-offs. One constraint is maintaining *market discipline*. For instance, allowing yields to rise due to market forces can impart a healthy signal about fiscal policy (high yields warning about deficits). If friendly actors always suppress yields, it could enable fiscal profligacy or create moral hazard for other investors (expecting a bailout). Thus, they must balance intervention with letting the market work. Another constraint is that any overt coordination (e.g. multiple central banks buying U.S. bills in concert) could be seen as improper or politically sensitive. They prefer subtlety. Moreover, institutional mandates constrain them: the Federal Reserve's mandate is domestic (maximize employment, stable prices) – while it cares about market functioning, it cannot openly say it's targeting Treasury yields except in extreme dysfunction. Foreign allies might support the U.S., but only up to a point that it aligns with their own reserve management needs. There's also the self-preservation constraint: even a friendly large holder (say Japan or a GCC country) won't accumulate infinite T-bills if it jeopardizes their portfolio (they have limits on exposure).

Expected system response: When friendly actors execute these stabilization strategies well, the system's response is typically a damping of volatility. For example, if a spike in yields is met by quiet but firm buying from aligned central banks, bid-ask spreads should tighten and yields settle, perhaps even reversing the spike. Because these actors are seen as "strong hands," their involvement often restores confidence; other investors may start buying once they perceive an official support. Ideally, the market normalizes without ever knowing exactly who intervened—at most, weeks later data might show an uptick in official holdings. The trade-off is that if used excessively, such support could lead investors to under-price risk (assuming "the Fed/alliances will always bail us out"), potentially sowing seeds for future leverage build-up. But in immediate terms, aligned intervention would enlarge the basin of attraction around a stable low-yield equilibrium (to use systems terms), making self-reinforcing sell-offs less likely. An example outcome: during a bout of stress, bill yields that might have shot up 100 bps perhaps only rise 20 bps and quickly fall back, with market functioning preserved (auctions still well-covered, repo markets stable). The friendly actors would then discreetly step back once confidence is restored.

### *B. Neutral/Self-Interested Actors*

These include major reserve managers from countries without explicit strategic alignment, sovereign wealth funds, global asset managers, hedge funds, etc. They are not looking to either prop up or undermine the U.S. per se; rather, they seek to optimize their portfolios under given constraints (regulatory, risk, return, liquidity). They respond to incentives and risks in a largely apolitical way.

Objectives: Primarily, preserve capital and earn a reasonable return. This group wants to ensure liquidity for their own needs (e.g. a reserve manager needs to know funds are accessible when needed), manage currency and duration exposures prudently, and adjust allocations as fundamentals change. They do not want to be caught holding an underperforming asset if better alternatives exist, but they also do not want to spark instability that could harm their own investments. Essentially, their aim is to avoid losses and maybe capture gains by reallocating when shifts in risk/reward arise.

Tools/Strategies: A common strategy here is *incremental diversification*. Over time, these actors might slowly rebalance away from Treasuries (including T-bills) if, for example, yields are unattractively low or U.S. fiscal metrics deteriorate. This "slow bleed" reallocation is done in small continuous flows to avoid moving the market—e.g., a central bank might reduce its T-bill holdings by a modest amount each month, reinvesting into other assets (gold, other countries' bonds) to diversify reserves. Neutral actors also manage currency risk: *FX-hedged vs. unhedged exits* matter. If a foreign fund wants to reduce Treasuries, they could sell U.S. bills and convert dollars to their home currency (unhedged exit), which puts downward pressure on USD. Alternatively, they might remain in USD but shift into dollar bank deposits or agency securities, etc., which changes the composition of dollar assets but not currency exposure. Many reserve managers hedge currency via forward contracts; a sign of gradual exit might be increasing forward sales of USD (which can widen cross-currency basis spreads). These decisions are made by comparing costs: e.g., if hedging USD yields eats the interest differential, then unhedged local-currency investments might become preferable.

Another set of strategies revolves around the pace of reallocation: *"slow bleed" vs. "event-driven" moves*. Under normal conditions, the slow bleed (gradual, predictable) is preferred to minimize market impact. However, if a specific event triggers alarm (say a ratings downgrade of U.S. debt or a sudden geopolitical shock), these actors may turn into *event-driven sellers*. For instance, if inflation surges unexpectedly and they fear the Fed will hike rates aggressively, asset managers might swiftly shorten duration by moving from notes to bills (which is a relative shift within Treasuries) or even from bills to pure cash if they think yields will rise further. Conversely, in a risk-off event, they might shift from other assets into T-bills (a flight to quality), thus *buying* bills en masse, as seen in crises when global funds park money in U.S. bills for safety.

A crucial element is that individually rational decisions can sum to collective outcomes: if many neutral actors independently decide to trim Treasury exposure by a small percentage (perhaps due to rising U.S. debt or better yields elsewhere), the aggregate result can be a significant outflow from T-bills. No single actor intends to crash the market, but their collective action can cause a drift upward in yields and stress in auctions. These actors watch the same indicators (Fed policy, U.S. fiscal outlook, liquidity conditions) and may herd somewhat in response to them, even without coordination.

Constraints: Neutral actors are constrained by their mandates and risk management rules. A central bank reserve manager, for example, must maintain a certain level of liquidity and safety—so they can't allocate all away from Treasuries without raising credit or liquidity risk elsewhere. They often adhere to conservative guidelines (e.g., X% must be in AAA sovereign debt, etc.), which anchors a baseline demand for T-bills. Sovereign wealth funds have strategic asset allocation targets that change slowly. Large asset managers are constrained by benchmark indices (many global bond indices heavily weight U.S. Treasuries), so even if they are bearish they can't zero out Treasuries without deviating from benchmarks (which brings career/alpha risk). Additionally, coordination among them is low (by design, to avoid anti-trust issues), so unlike a single minded adversary, neutral actors rarely act in a concerted simultaneous way—they are more diffuse. They also face the *first-mover problem*: if one starts selling heavily and pushing yields up, others might suffer mark-to-market losses on their remaining holdings—so there is some implicit incentive not to be the one that sparks a rout. In practice, many will try to shade down exposure without rocking the boat.

Expected system response: The actions of neutral actors tend to manifest as gradual trends or mild pressure points rather than sudden dislocations—until they possibly crescendo at a trigger point. For instance, if over a quarter, several central banks each reduce their bill holdings a bit, the net effect could be a perceptible uptick in T-bill yields relative to policy rates, or a slight decline in bid-to-cover ratios at bill auctions. The market usually can digest a slow bleed if it's within historical norms. Often the system responds via price adjustments: yields inch up to entice other buyers (like domestic money funds) to pick up the slack, reaching a new equilibrium. In stable times, these marginal adjustments don't impair functioning; they just reallocate who holds the bills (e.g., foreigners sell, U.S. mutual funds buy).

However, if an event-driven sell-off by neutrals occurs, the immediate response may be a sharp but short-lived jump in yields and decline in liquidity. For example, a surprise hawkish Fed meeting might cause global funds to dump short-term Treasuries, pushing bill yields up swiftly by, say, 20-30 bps in days. The system's response would then depend on aligned actors or value-driven private buyers stepping in at the higher yields. Often, such moves correct somewhat: once yields are higher, some neutral actors will actually step back in (those who were underweight Treasuries might see value at the new yields). So neutral-driven volatility tends to find an equilibrium after overshooting a bit. Importantly, because these actors are not trying to break the system, if conditions stabilize or their models show Treasuries as cheap, they will supply liquidity by buying back in. In summary, neutral actors can amplify cycles (buying more when yields fall, selling when yields rise – i.e., momentum or pro-cyclical behavior), but they also provide depth to the market under normal conditions. The net systemic effect is that they can cumulatively create drift or mild waves in yields, occasionally a squall, but rarely a full storm on their own.

### C. Hostile/Adversarial Actors

This category comprises actors who might seek to use financial channels to weaken U.S. economic or geopolitical standing. Examples often cited include certain sanctioned states or great-power rivals holding large dollar reserves who could, in theory, dump Treasuries to cause disruption. It could also include non-state actors (large activist funds) aligning with a view to short U.S. debt, though in practice state actors are the typical concern. These actors might be willing to suffer economic costs to themselves in order to inflict stress on the U.S., making their behavior less constrained by pure profit motives.

Objectives: A hostile actor's goals could include: (1) *Liquidity disruption* – deliberately causing a shortage or surge in T-bill liquidity to spark turmoil in funding markets. For example, suddenly selling a large quantity of T-bills could drive yields up and perhaps cause ripple effects in money market rates and confidence. (2) *Political signaling or leverage* – using the threat or act of Treasury sales to send a message (e.g., as retaliation for sanctions or other policies) or to pressure U.S. policymakers. This has been colloquially described as the "nuclear option" of dumping U.S. debt [8]. (3) *Raising U.S. fiscal/inflationary constraints* – by pushing up yields, they increase U.S. borrowing costs, potentially constraining fiscal space or forcing the Federal Reserve into difficult choices (for instance, if the Fed intervenes to cap yields, it might undermine its inflation-fighting credibility, etc.). In essence, the adversary might aim to increase financial stress in the U.S., contribute to domestic economic difficulties, or undermine global confidence in U.S. leadership.

Tools/Strategies: A hostile actor with a large T-bill (or overall Treasury) portfolio can engage in *coordinated selling*. This might involve not just the primary actor but allied adversaries or proxy institutions acting in concert to unload Treasuries in a short span. For example, Country A and Country B (unfriendly to the U.S.) could both dump, say, $50 billion of T-bills over a week, hoping to flood the market. Such selling could be in the secondary market, putting pressure on prices, and also by not rolling over at auctions (i.e., boycotting or submitting punitive high-yield bids to auctions to make the Treasury's financing more expensive). Another tactic is *auction disruption*: an adversary could put in outsized competitive bids at auction at much higher yields (or conversely, abstain entirely), aiming to produce a tail (the auction clearing yield coming in much higher than expected) which could alarm markets about Treasury demand. In extreme forms, one could envision an adversary trying to engineer a failed auction by rallying others not to bid—though the primary dealer system and broad auction participation make a total failure unlikely.

They could also exploit the *repo and collateral channel*. For instance, an adversary could withdraw a large amount of Treasuries from the repo market (if they typically lend out their holdings). By refusing to roll their repos or pulling collateral, they might create scarcity that sends repo rates soaring, as happened in smaller scale in Sept 2019 for technical reasons [6]. Conversely, they could lend out a glut of Treasuries (if they have them idle) to push repo rates down and drain dealer balance sheets. Another angle is using derivatives or dollar funding markets to amplify stress: e.g., short-selling T-bill futures or using FX swaps markets to make USD suddenly scarce, forcing others to dump Treasuries for cash.

A hostile strategy might also be timed to strike at vulnerable moments. For example, doing this during a debt ceiling impasse or during a Fed tightening cycle when liquidity is already thinner could maximize impact. They might also engage in *information operations* – spreading rumors or public statements about "dumping U.S. debt" to rattle markets (even without immediate action, jawboning can cause investors to preemptively sell). Indeed, China's potential to "weaponize" its Treasury holdings has been periodically floated in media [8], even if not acted upon. An adversary could quietly shorten the maturity of their holdings (switching from notes to bills as a temporary tactic [8]) to signal lack of confidence in long-term U.S. finances – this wouldn't crash the market but could nudge long-term yields up and telegraph a negative outlook.

Constraints: Despite the fears, adversaries face significant self-imposed constraints when manipulating T-bills. Firstly, *self-harm* is a major deterrent: dumping Treasuries would "crush the value of [their] own holdings" [8], inflicting capital losses. A large-scale sale would be quickly reflected in price declines, so they'd be shooting themselves in the foot while trying to poke the U.S. in the eye. If they hold hundreds of billions, even a 1% drop in prices is a big loss to their reserves. Secondly, markets anticipate such moves; a whiff of a big seller can lead others to front-run the sale, exacerbating the adversary's losses [8]. There is also the near-certain *policy response*: the Federal Reserve (possibly in coordination with allies) would likely respond to disorderly conditions with large-scale purchases, blunting the impact [8]. An adversary cannot reliably count on breaking the market faster than the Fed can fix it, especially given the Fed's essentially infinite dollar liquidity capacity. Another constraint is that Treasuries, especially T-bills, are held for good reasons by these actors (liquidity, safety of reserves). If they sell them, they must place the proceeds somewhere—likely into other dollar assets or gold or such. Some alternatives, like gold, are far less liquid and could spike in price (making their entry costly). Others, like other countries' bonds, are not deep enough to absorb a huge shift (the U.S. Treasury market is unique in size). Furthermore, a hostile dump would invite *retaliation*: the U.S. could respond with financial sanctions, freezing assets, or other measures that leave the adversary worse off. Knowing this, truly hostile actions with Treasuries have been scant historically. Even during periods of tension, major holders have generally opted for gradual reduction rather than sudden liquidation.

There's also a coordination problem on the adversary side: if multiple adversaries exist, they may not fully trust each other to act in unison (one might defect or quietly hedge). Unilateral action by one adversary might just allow others (even if adversarial in intent) to buy what's sold at cheaper prices—thus one could undercut the other.

Expected system response: If an adversary attempted a significant hostile manipulation, the immediate effect could be a spike in yields and a hit to market confidence. For instance, suppose a large holder sold $100 billion of T-bills in a short period. Yields on those maturities might jump noticeably (tens of basis points) and market liquidity could suffer as dealers absorb the influx. We might see disorderly trading—widened bid-ask spreads, a surge in repo fails if the collateral flows are disrupted, and possibly a spillover into the FX market (a large sale with dollar conversion could weaken the USD temporarily). However, the U.S. authorities and aligned actors would likely react swiftly. The Fed could announce open market purchases or repo operations to stabilize money markets (essentially taking the other side of the trade). In modern times, automated trading and circuit-breakers in futures might also slow a crash. It's plausible that the system would interpret a one-off hostile dump as an arbitrage opportunity: many investors would believe the Fed or others will step in, so they might buy the dip in Treasuries. In effect, the adversary's Treasuries get redistributed to other holders at a discount, and after some volatility, yields settle perhaps only modestly higher than before.

That said, repeated or sustained hostile actions (a campaign of selling) could raise risk premia. The market might start pricing a "security premium" if Treasuries are seen as a geopolitical pawn—for example, slightly higher yields to compensate for potential future disruptions. In a extreme scenario where an adversary's actions coincide with other stresses (say during a U.S. domestic crisis or global recession), it could contribute to an unstable equilibrium where confidence is shaken. But in general, because of the constraints mentioned, hostile actors have so far mostly used threats rather than follow through. China, for example, has steadily reduced its relative holdings over years (from ˜14% of Treasuries a decade ago to <3% now) [8], but in a gradual way that did not crash the market. Analysts note that outright weaponization of Treasuries remains unlikely [8] [8], precisely due to these feedbacks and self-goals.

In summary, an adversarial attempt to use T-bills as a weapon could at most create short-term dislocations that the system (via the Fed and other investors) is equipped to counter. It would send political signals and perhaps marginally increase U.S. borrowing costs for a time, but it is an unstable strategy. Over the long term, such actions would probably prompt changes in U.S. debt management (e.g., more issuance in other maturities or encouraging a more dispersed holder base) and international coordination to fortify the Treasury market's resilience.

## III. Deep Dive: State Variables, Indicators & Equilibrium Models

We turn to a structural analysis of the T-bill market as a dynamic system. Here we identify the key *latent state variables* that T-bills reflect or influence, the observable *indicators* that serve as proxies or signals for those states, and conceptual *equilibrium models* describing the system's behavior. We also discuss how feedback loops can lead to multiple equilibria or regime shifts, and what factors determine system stability.

### A. Hidden State Being Manipulated

At the core, T-bill markets encapsulate a **latent state of systemic liquidity and confidence**. This "state" is not directly observed but can be thought of as the aggregate degree of trust and ease in the dollar funding ecosystem. Several conceptual components of this state include:

- **Dollar Liquidity Stress Level ($\Lambda$):** A measure of how tight or loose dollar funding is globally. When $\Lambda$ is high (meaning stress is high), cash is scarce relative to demand – typically T-bill yields would be elevated above expected policy rates as investors demand yield for parting with cash (or conversely, in some crises, bill yields can drop below zero if the scramble is for safe assets; it depends on whether the dash-for-cash leads to selling of bills or buying of bills as safe haven, which can vary by phase of crisis). $\Lambda$ influences cross-currency funding (like FX swap basis) and repo rates as well.
- **Confidence in Fiscal Rollover ($\Psi$):** The perceived probability that the U.S. can roll over its debts without incident. This is especially pertinent to bills since they roll frequently. If $\Psi$ drops (say due to debt-ceiling concerns or fear of a technical default), the hidden state tilts towards risk-off: bill yields may diverge from other short rates as a default premium creeps in [5]. $\Psi$ is essentially the market's trust in the U.S. Treasury's near-term creditworthiness and payment logistics.
- **Market-Making Capacity ($M$):** This reflects the ability and willingness of dealers and other liquidity providers to intermediate trades. It's determined by factors like dealer balance sheet space, regulatory leverage ratios, and risk appetite. A healthy $M$ means any selling of T-bills finds buyers with little price impact (tight bid-ask, deep order books). If $M$ shrinks (dealers pull back), the market's state worsens: even moderate order flow can cause outsized yield moves. Market-making capacity was a limiting factor in March 2020's turmoil [3] when dealers hit constraints and couldn't absorb sales, until the Fed stepped in.
- **Credibility of Backstops ($B$):** The market's confidence that authorities (primarily the Fed, but also Treasury via tools like the Exchange Stabilization Fund or standing swap lines internationally) will step in to stabilize things if needed. If $B$ is high, participants are less likely to panic sell or hoard liquidity—knowing, for example, that the Fed's standing repo facility will provide cash against Treasuries in stress. If $B$ is in doubt (perhaps due to policy constraints or political delays), the system is more fragile. An example: if inflation is 10% and the Fed is hesitant to buy Treasuries, markets might question the backstop, effectively lowering $B$.
- **Inflation vs. Financial-Stability Constraint Balance ($\Theta$):** This variable represents the trade-off or tension in policy priorities between controlling inflation and ensuring market stability. It's not an observable metric but a contextual state. When $\Theta$ tilts toward inflation-fighting, market participants know the Fed might tolerate more financial volatility (i.e., let yields rise, not intervene immediately). This might raise risk premia in bills because the safety net is a bit further out. Conversely, if $\Theta$ tilts toward stability (for instance, if inflation is low or the Fed explicitly signals worry about market functioning), the implied put for T-bills is stronger, compressing risk spreads. Essentially, $\Theta$ influences how the hidden state responds to shocks: e.g., in late 2022, with high inflation, a liquidity stress might persist longer because the Fed's hands are semi-tied – a different regime than 2020 when inflation was low and the Fed could immediately flood markets with liquidity.

| Actor | Objectives | Tools/Strategies | Constraints | Expected System Response |
|---|---|---|---|---|
| *Friendly/Aligned* (e.g. U.S. allies, Fed, banks) | Maintain stability; smooth functioning; support U.S. financing | Portfolio smoothing (buy on weakness); maturity laddering; coordinated reinvestment; confidence signaling | Preserve market discipline (avoid moral hazard); limited by mandates; informal coordination | Volatility dampens; yields revert to equilibrium; auctions covered. Risk of moral hazard if overused. |
| *Neutral/Self-interested* (e.g. reserve managers, funds) | Safeguard value; ensure liquidity; optimize returns | Gradual diversification; FX-hedged vs unhedged shifts; flight-to-quality in crises; event-driven reallocations | Bound by investment guidelines/benchmarks; need liquid reserves; herding risk | Gradual shifts cause mild yield trends. In shocks, yields move sharply but find new equilibrium. Generally mean-reverting. |
| *Hostile/Adversarial* (e.g. sanctioned or rival states) | Undermine U.S. flexibility; signal displeasure; induce instability | Coordinated large sales; boycott/bid up yields at auctions; repo manipulation; public threats | Self-inflicted losses; Fed/others counteract; limited alternatives; risk of retaliation | Initial yield spike and turmoil. Fed intervenes, allies buy dip – market stabilizes. Could raise risk premia if repeated. |

TABLE II

*Mapping of actor classes to their objectives, tools in T-bill markets, constraints, and likely system responses.*

One can think of the overall hidden state $X$ as a function of these components: e.g. $X = f(\Lambda, \Psi, M, B, \Theta)$ that summarizes "systemic liquidity and confidence." T-bill markets both reflect $X$ (through prices and flows) and can influence $X$ (because if T-bill market freezes, that itself reduces $\Lambda$ and $M$ further, etc.). Global actors manipulating T-bills are effectively trying to nudge this hidden state: adversaries may try to push $X$ into a "stressed" zone, whereas authorities try to keep $X$ in a normal zone.

### *B. Indicators & Observables*

While $X$ is latent, a rich set of observable indicators can proxy for different aspects of it. We enumerate key indicators and classify them by their relationship to underlying state, timing (leading, coincident, lagging), and whether they're structural or cyclical signals.

- **T-Bill Yields vs. Policy Rates:** The spread between T-bill yields (say the 3-month bill rate) and the Federal Reserve's policy rate (or OIS of corresponding maturity) is a crucial barometer. In normal times, 3m T-bills might trade a few basis points below the overnight index swap (OIS) rate due to their liquidity premium. If bills trade significantly *above* OIS or the Fed's Interest on Reserve rate, it indicates stress – investors demanding extra yield (cash shortage or credit concern). If bills trade far *below* OIS (even negative yields while policy rate is positive), it indicates an acute demand for safety or collateral, pushing yields down (as seen at times when MMFs poured into bills and the Fed's RRP) [4] [3]. This spread is a leading/-coincident indicator: it often widens ahead of or during stress (e.g., before quarter-ends if liquidity is forecasted to tighten, bills cheapen). It's relatively cyclical, fluctuating with liquidity conditions, though persistent deviation can signal structural changes (e.g., a secular shortage of safe assets). It's fairly sensitive (not too noisy) but regime-dependent (its interpretation can flip if, say, Fed policy itself is in transition).
- **Auction Metrics (Bid-to-Cover, Tail, Allotment by Investor Type):** Every T-bill auction provides real-time data on demand. Bid-to-cover ratio (total bids divided by amount offered) measures the breadth of demand; a ratio consistently dropping below historical norms (e.g., under 2.5 for 10-year notes, or analogous levels for bills) is an early warning [7]. The auction "tail" – the difference between the auction stop-out yield and the when-issued yield just before the auction – indicates whether demand was weaker or stronger than expected [7]. A significant tail (auction yield higher than market expected) is a coincident indicator of demand shortfall and often briefly rattles secondary markets. Participation metrics (what fraction was bought by primary dealers vs indirect bidders like foreign central banks) can hint at structural shifts: e.g., if indirect (foreign) bid declines over several auctions, that's a structural change in demand composition. Auction metrics are leading indicators in the sense that a poor auction can trigger broader sell-offs (it's a stress test of the market's appetite). They're relatively high-frequency and noisy (one weak auction can be an outlier), so one looks for trends. They straddle structural/cyclical: a secular decline in bid-to-cover could mean structural waning of demand, while an occasional low bid-to-cover might be just cyclical timing issues.
- **Repo Rates and Fails, Haircuts:** The repo market provides daily insight into the money-market functioning of Treasuries. The Secured Overnight Financing Rate (SOFR) or Treasury GC (general collateral) repo rate compared to the Fed's rate is a key indicator. If repo rates spike above the Fed's corridor (as in Sept 2019) [6], it flags a cash scarcity or imbalance in collateral demand. Persistent elevation of repo rates relative to bills might mean that holdings are strained (people paying more for cash). Conversely, ultra-low repo rates (even below the Fed's RRP rate floor) indicate excess cash or scarce collateral (people eager to lend cash against Treasuries, possibly due to needing collateral). Repo *fails to deliver* – when a party can't deliver Treasuries in a repo transaction – often jump when specific CUSIPs are in short supply

$$X = f(\Lambda, \Psi, M, B, \Theta)$$

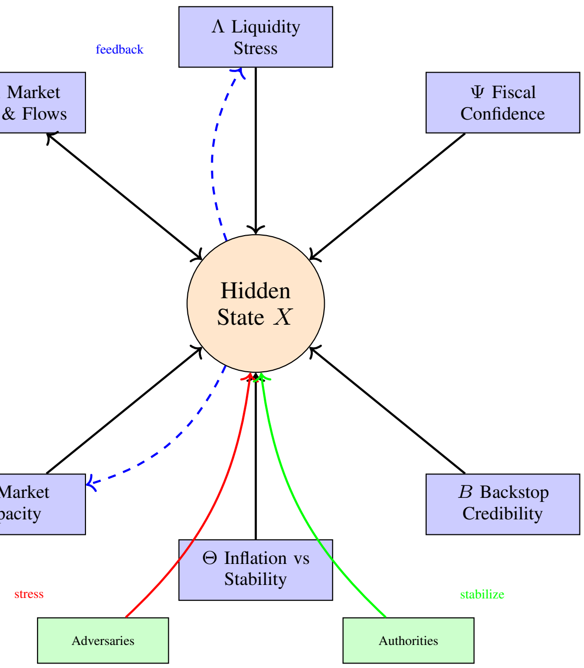


Fig. 3. *Hidden State Variables Model. The overall system state X is a function of five key components: Dollar Liquidity Stress ($\Lambda$), Confidence in Fiscal Rollover ($\Psi$), Market-Making Capacity ($M$), Credibility of Backstops ($B$), and Inflation vs. Stability Balance ($\Theta$). T-bill markets both reflect and influence this hidden state, while external actors seek to manipulate it.*

or if there's market stress (since fails are tolerated instead of delivering at high cost). High fails are a sign of collateral stress (often coincident with liquidity issues). Haircuts (the margin required on Treasuries as collateral) are usually stable for T-bills (very low haircuts), but in extreme stress, lenders might demand higher haircuts even on bills (a lagging indicator since haircuts usually change after sustained volatility). Repo metrics are more *technical coincident indicators* – they reflect the plumbing stress directly as it happens, often spiking intra-day. They are cyclical, though extreme behavior might point to structural leverage issues. They can be noisy day-to-day (due to calendar effects, etc.), but a trend of rising repo rates or increasing fails outside known technical periods is a red flag.

- **Market Liquidity Measures (Bid-Ask Spreads, Market Depth):** These directly gauge $M$, the market-making capacity. In normal times, T-bills have very tight bid-ask spreads (fractions of a basis point) given their liquidity. If market-makers withdraw, the bid-ask spread widens and quoted depth at the top of the book falls. Depth can be measured by the quantity available for sale/purchase at the best quotes on electronic interdealer platforms. During March 2020, for example, depth on benchmark Treasuries plummeted and bid-ask spreads widened multiple-fold, indicating a loss of liquidity [3]. For bills, data might not be as readily published, but primary dealers and brokers report such conditions qualitatively. This is a coincident indicator of stress: when liquidity evaporates, it's happening in the moment. It's also a good regime-sensitive indicator: in normal regime, it's stable and low; in crisis regime, it shifts dramatically. However, it can be volatile (noisy) at high frequency and often requires high-frequency data to track. Still, a persistent change (say bid-ask that used to be 1/64th now routinely 3/64ths) may signal a lasting structural reduction in liquidity provision, perhaps due to regulations or dealer retrenchment.
- **Money Market Fund Flows & Yields:** Money market funds (MMFs) are huge players in T-bills (holding ~20% of the market) [4]. Their flows can serve as a gauge of cash moving in or out of the T-bill space. For instance, in times of stress, investors often reallocate from prime funds (which hold some credit assets) to government funds (which hold Treasuries and repos), causing large inflows into government MMFs which then demand more T-bills. Such flows were seen in March 2020, when government MMFs saw big inflows (seeking safety) even as prime funds had outflows. Conversely, if MMFs start experiencing outflows (perhaps because yields elsewhere become more attractive or due to policy changes), they will reduce T-bill purchases. Weekly MMF AUM changes, provided by agencies like the Investment Company Institute, are leading or coincident indicators: rapid inflows into government funds often coincide with

early flight-to-quality, pushing bill yields down (or preventing them from rising). Outflows might lead to selling of bills or at least less demand at rollovers, putting upward pressure on yields. MMF yields (the net yields they pay investors) also indicate if T-bill yields are transmitting to end-investors; if there's a divergence (say T-bill yields rise but MMF yields lag due to portfolio WAM, etc.), it might temporarily influence investor behavior (moving money for better yield). Overall, MMF data are somewhat structural (they depend on regulations like liquidity buffers and Fed's RRP facility usage) yet also very cyclical (responding to Fed rate changes and risk sentiment).

- **FX Market Behavior Conditional on Yield Moves:** The USD exchange rate's co-movement with Treasury yields can offer clues about foreign flows. Typically, rising U.S. yields relative to other countries attract capital, tending to strengthen the dollar (all else equal). If we observe a scenario where U.S. T-bill yields are rising but the dollar is *weakening*, it suggests that foreigners might be selling U.S. assets (pushing yields up) and converting proceeds out of USD (pushing USD down). Such a divergence occurred in certain historical episodes and can be a leading indicator of foreign liquidation. It is regime-sensitive: in a "risk-on" regime, yields up & USD up might just mean optimism about U.S. growth; in a "stress" regime, yields up & USD down could mean foreign dumping (or concern about U.S.-specific risk such as twin deficits). Conversely, if bill yields fall while USD strengthens, it could indicate a rush into dollar liquidity (people buying dollars and stashing into bills). So, looking at correlations between short-rate changes and dollar index moves can help infer flow direction. This indicator is a bit noisier as FX is influenced by many factors (trade, Fed expectations, etc.), but abnormal correlation patterns (breaking of usual correlations) often accompany regime shifts or unique shocks.

In addition to these, other indicators worth tracking include: *Treasury yield curve shape at the very front end* (e.g., 1-month vs 3-month bill yields – a kink might indicate debt ceiling risk localized to a certain maturity), *Cross-currency basis spreads* (cost to swap other currency into USD – widens when USD funding stress rises, often paralleling bill market tightness), and *OIS–Treasury spread* (similar to above but from another angle of risk premium). Each indicator can be classified as:

- **Leading**: tends to move or flash warning before the broad stress is recognized. E.g., a widening FX swap basis or subtle decline in indirect bids at auctions might precede broader volatility.
- **Coincident**: moves essentially as the event unfolds. E.g., repo rate spikes and bid-ask spreads are coincident with acute stress.
- **Lagging**: change only after the fact. E.g., official data on foreign holdings or changes in MMF allocations come with a lag, confirming what happened.

And:

- **Structural**: those that change when there's a fundamental regime/equilibrium shift. E.g., a long-term decline in foreign ownership percentage, or a permanently higher average bid-ask due to regulations.
- **Cyclical**: those that oscillate with cycles but revert. E.g., bid-to-cover might oscillate around a mean depending on issuance sizes and risk sentiment, but if it returns to normal later, that was cyclical.
- **Noisy vs. Regime-sensitive**: Noisy indicators have lots of short-term volatility (like daily FX moves) which might not mean much unless a threshold is breached or sustained trend appears. Regime-sensitive means they behave very differently in crisis vs calm (e.g., repo fails might be near zero in normal times, but jump in crisis – effectively zero until they're not, which is regime change).

The art for an analyst or monitoring system is to use an *indicator cluster*. Rarely does one indicator alone confirm the hidden state shift – but if several start flashing (say: bill-OIS spread widening, repo rate elevated, FX basis widening, bid-to-cover falling all at once), one can be confident the system is moving to a stressed regime. Conversely, one-off noise (e.g., one weak auction) might not signify much if others remain normal.

### *C. Equilibrium & Feedback Models*

We can frame the T-bill market (and connected funding system) as a dynamical system that can exhibit multiple equilibria and nonlinear feedback effects. Let's discuss qualitatively first, then outline a simple quantitative stylized model.

Qualitative equilibrium concepts:

- The system potentially has **multiple equilibria**: a *stable, low-yield equilibrium* where confidence is high, liquidity is ample, and T-bills trade at tight spreads (a virtuous circle: everyone's comfortable holding bills, so yields stay low, so financing is easy, reinforcing comfort). And an *unstable, high-yield equilibrium* (or at least a metastable one) where confidence erodes, many actors want out, yields jump to attract reluctant buyers, and volatility is high – this could be self-fulfilling to a point (people sell because yields/spreads are rising, which in turn causes further rise). In extreme thought-experiment, a third equilibrium would be a failed state (if no one buys, yields approach prohibitive levels or auctions fail – but in reality, mechanisms kick in before that point). The presence of multiple equilibria is common in sovereign debt models: if investors believe others will flee, it's rational for each to flee (bad equilibrium); if they believe others will stay, each is content to stay (good equilibrium). The U.S. enjoys a strong bias toward the good equilibrium due to trust and Fed backstop, but near critical moments (like the debt ceiling), the market oscillated between these expectations.
- **Phase transitions:** The system can exhibit sudden shifts – e.g., from liquid to illiquid market in a matter of days ("phase change"). March 2020 is an example where the Treasury market went from one phase (normal function-

ing, yields reflecting fundamentals) to another phase (fire-sale, yields reflecting pure liquidity preference) almost overnight [3]. These transitions often happen when some threshold is breached – for instance, when selling volume overwhelms dealer capacity or when a critical mass of investors moves to hoard cash. Before such a transition, indicators might move gradually; at the transition, many indicators spike abruptly. An equilibrium that was stable becomes unstable beyond a tipping point.

- **Threshold effects:** Certain conditions create tipping points. For example, *forced sellers* like in the UK LDI crisis: if yields rose above X, margin calls forced more sales, which pushed yields further – a classic positive feedback loop. In T-bills, margin-driven feedback is less (since they are short duration and not typically margined like bonds), but other thresholds exist: e.g., if the Fed's RRP rate (floor) is X%, normally bill yields won't go much below that. But if they do (due to extreme demand), money funds might hit an arbitrage limit where they stop buying bills and put money in the RRP facility instead – that's a threshold that caps how low bill yields can go relative to policy in normal times. On the upside, regulatory balance sheet limits can act as thresholds: dealers might absorb selling until their capital utilization hits, say, 100% of a limit, at which point their demand falls off sharply and yields must jump to find another marginal buyer (like the Fed or a non-regulated entity). Another threshold: foreign central banks sometimes have internal mandates on how much to keep in USD assets; crossing a geopolitical threshold (like sanctions risk) might cause a bloc to significantly cut Treasuries.
- **Balance-sheet limits and feedback:** When banks/dealers reach leverage or liquidity coverage ratio limits, they stop intermediate smoothly, leading to a breakdown where normally a small rate move would clear the market, now a large move is needed. This is how a previously stable equilibrium (with small oscillations) becomes unstable beyond a point – essentially the *Jacobian* of the system changes sign for certain variables, from negative feedback to positive feedback. For instance, under normal conditions, if yields rise a bit, value investors come in (negative feedback – pushes yields back down). But if all value investors are full up or constrained, a rise in yields doesn't attract sufficient new buyers, and may actually cause others to sell (those who fear further rise), thus yields have to rise more – a positive feedback regime.

Now, a simplified **quantitative model** to illustrate equilibrium and Jacobian stability:

Consider a supply-demand framework for T-bills. Let $P$ be the price of a T-bill (so yield $y \approx \frac{1}{P} - 1$ for simplicity of continuous compounding near par). *Supply* $S$ of T-bills is set exogenously by the Treasury (though over long term, it's adjusted, here treat short-term as fixed stock to roll). *Demand* $D(P, X)$ depends on price and the state variables $X$ (like liquidity preference). In a stable situation, demand for T-bills is high when price is high (yield low) because people want safety (i.e., an inelastic demand that still clears at low yield). However, we can invert to think in yield terms, demand as function of yield $y$: normally $D(y)$ slopes downward (higher yield $\rightarrow$ more demand, as Treasuries become more attractive). But beyond a point, if $X$ deteriorates (loss of confidence), the demand curve might steepen or even bend backward for a time – e.g., if yields start rising fast, some holders may exit (so demand at a given yield actually falls because of panic).

We could formalize a simple piecewise demand:

$$D(y; \Psi, \Lambda) = A(\Psi, \Lambda) - B \cdot y + C \cdot f(\Lambda, y) \tag{1}$$

Where $A$ is base demand influenced by confidence $\Psi$ and liquidity $\Lambda$, $B$ is the normal elasticity (positive number, investors buy more if yield higher), and $C \cdot f(\Lambda, y)$ captures aberrant effects: say if liquidity stress $\Lambda$ is high, $f$ might be an increasing function of $y$ (meaning as yields rise, $f$ adds to supply instead of demand if investors are forced to sell). In normal times $\Lambda$ is low so $f(\Lambda, y) \approx 0$ and demand is nicely downward sloping. In stress, $\Lambda$ high, maybe $f$ kicks in such that for $y > y^*$, $f$ becomes positive (net reduction in demand as yields rise beyond threshold $y^*$ because of forced selling or fear).

Equilibrium yield $y^*$ is where $D(y^*) = S$ (demand equals fixed supply). If multiple intersections, the stable one is where demand curve crosses supply with a negative slope (so small deviations bring it back) and unstable one if positive slope.

The **Jacobian stability** in a dynamic sense: Imagine a dynamic adjustment

$$\dot{y} = \alpha[S - D(y)], \tag{2}$$

meaning if demand is less than supply (excess supply), yields rise ($\dot{y} > 0$) to clear; if demand exceeds supply, yields fall. Linearizing around equilibrium $y^*$,

$$\dot{y} \approx -\alpha D'(y^*)[y - y^*]. \tag{3}$$

The sign of $D'(y^*)$ (the slope of demand curve at equilibrium) determines stability. Normally $D'(y^*) = -B < 0$ (downward sloping demand), so $-\alpha D' > 0$. The equilibrium is stable if the coefficient on $(y - y^*)$ is negative (so if $y$ is above $y^*$, $\dot{y}$ is negative to pull it down). That requires $-\alpha D' < 0 \implies D'(y^*) > 0$.

Let me be careful with the stability condition. In price terms, if $\dot{P} = \beta[D(P) - S]$ (if demand > supply, price rises), then at equilibrium $P^*$, linearize: $\dot{P} \approx \beta D'(P^*)(P - P^*)$. Since $D'(P) < 0$ normally (higher price $\rightarrow$ lower demand), we have $\beta D' < 0$: $\dot{P} \approx$ (neg coeff)$(P - P^*)$. This is stable: if $P$ above eq, $\dot{P}$ negative (falls back); if $P$ below, $\dot{P}$ positive (rises). Good. So indeed $D'(P) < 0$ (downward sloping in price axis) yields stability.

In yield terms, that corresponds to stability when the demand curve slopes down in yield space (higher yields attract more demand, i.e., $D'(y) > 0$ normally for Treasuries). The equilibrium becomes unstable if heavy positive feedback kicks in (like more selling as yields rise), causing the system to jump to a new equilibrium.

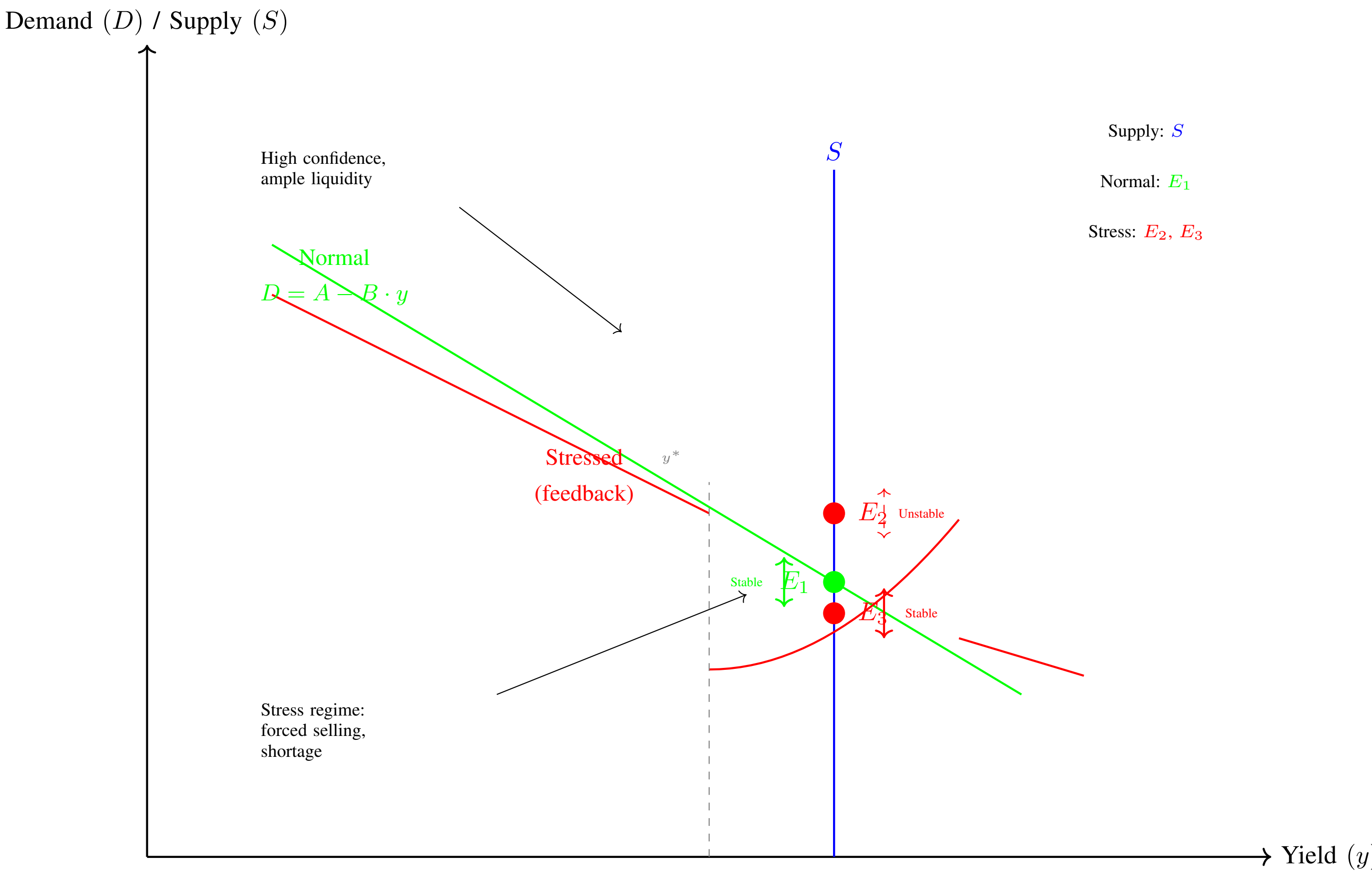


Fig. 4. *Supply-Demand Equilibrium with Multiple Equilibria. In normal conditions (green), demand slopes downward and intersects supply at stable equilibrium $E_1$. Under stress (red), the demand curve can bend backward due to forced selling beyond threshold $y^*$, creating multiple equilibria: unstable $E_2$ and stable high-yield $E_3$. The system can jump between equilibria depending on market conditions.*

Conditions shifting equilibria: Changes in state parameters like $\Psi$ (credit confidence) or $\Lambda$ (liquidity stress) effectively shift the demand curve. For instance, if $\Psi$ (rollover confidence) drops, the entire demand at a given yield might decrease (investors require a higher yield to hold bills) – equilibrium yield $y^*$ moves higher. If $\Lambda$ increases (stress), part of demand might turn into net supply from forced selling beyond a threshold – possibly eliminating the low-yield equilibrium entirely, leaving only a higher-yield equilibrium. Graphically, the supply-demand intersection might jump from one branch to another. This is a typical bifurcation in system dynamics.

Widening/narrowing basins of attraction: The basin of attraction is how far the system can be perturbed and still return to original equilibrium. In calm times with strong backstops ($B$ high) and ample dealer capacity ($M$ high), even a large temporary sell-off will be absorbed and the system falls back to the prior equilibrium (basin wide). But if those stabilizers weaken, a much smaller shock could push the system past the tipping point into the other basin (basin of normal eq narrows). For example, if dealers are near limits and the Fed is on the sidelines, a moderate sell pressure might not get countered, and yields escalate until a new clearing price is found (which could be significantly different).

Policy intervention and topology changes: When the Fed or Treasury intervene (like establishing a new facility), it can alter the system's topology—essentially removing or smoothing out an unstable region. One can view, say, the Fed's standing repo facility (SRF) as adding a quasi-infinite demand for Treasuries at a certain repo rate, which translates to a floor on Treasury prices in stress. This can eliminate an otherwise runaway feedback loop where repo rates spike and force sales. Another example: during COVID, the Fed's massive QE basically guaranteed demand at near-par for a huge volume of Treasuries, cutting off the tail of potential equilibria where Treasuries would trade at fire-sale prices. By doing so, they ensured the system had a single equilibrium (the stable one). In technical terms, policy can add a strongly negative feedback (if yields rise too much, Fed buys, pushing them down) which makes the Jacobian's eigenvalues negative again (restoring stability).

We can also cast a simple **dynamical system example** with two state variables, to illustrate Jacobian eigenvalues: Let $x_1 =$ deviation of T-bill yield from normal, $x_2 =$ a measure of liquidity stress. Suppose:

$$\dot{x}_1 = a\,x_2 - b\,x_1 \tag{4}$$

$$\dot{x}_2 = c\,x_1 - d\,x_2 \tag{5}$$

Here $a, b, c, d$ are positive constants. The term $ax_2$ means yields go up if stress rises (liquidity stress pushes yields higher), and $-bx_1$ is a stabilizing term that if yields are above normal, eventually mean-reversion pulls them down (maybe through value buyers). The second eq: $cx_1$ means stress increases if yields rise (for instance, higher yields can indicate or cause funding issues), and $-dx_2$ means stress tends to dissipate over time if not constantly fueled (like liquidity eventually returns). The equilibrium is at $(x_1 = 0, x_2 = 0)$ i.e., normal conditions. The Jacobian matrix is

$$J = \begin{pmatrix} -b & a \\ c & -d \end{pmatrix}. \tag{6}$$

The eigenvalues satisfy $\lambda^2+(b+d)\lambda+(bd-ac) = 0$. Stability requires both eigenvalues to have negative real parts, which (for a 2x2 with real coeffs) means $\mathrm{Tr}(J) = -(b + d) < 0$ (which it is), and $\mathrm{Det}(J) = (bd - ac) > 0$. The determinant condition $bd > ac$ is key: if $bd > ac$, the system is stable (both eigenvalues negative). If $ac > bd$, one eigenvalue becomes positive – an unstable equilibrium emerges.

Here $ac$ can be thought of as the product of the cross-feedback (yield $\rightarrow$ stress via $c$, stress $\rightarrow$ yield via $a$), and $bd$ the product of self-correcting forces. So if the feedback loops ($a$ and $c$) overpower the dampeners ($b$ and $d$), the system tips into instability. In our context: $a$ might become large if suddenly stress strongly pushes yields (e.g., panic selling effect), and $c$ large if yields spur more stress (e.g., through mark-to-market losses or funding costs). If the Fed's backstop is credible, $b$ (yield mean reversion) is large, as any yield rise is quickly countered. If dealer/liquidity recovery is quick, $d$ is large (stress decays fast). So policy can effectively increase $b$ and $d$, ensuring $bd \gg ac$. Remove backstops, $b$ small, and if panic sets in, $a, c$ large, then $ac > bd$ and the equilibrium at low yields collapses—yields shoot up until some nonlinear limit maybe creates a new equilibrium.

Multiple equilibria and tipping: If unstable, the trajectories might not diverge to infinity but could converge to a different stable point (if any). In non-linear real world, yields won't go infinite; instead, another equilibrium might be where, for example, yields have risen enough that completely new buyers step in (like value investors or other central banks) to establish a new stable point, albeit at higher yields and with maybe different participants. That's the "runaway dynamic" until a break: e.g., yields keep rising until the Fed finally steps in at some pain threshold, which then becomes effectively the new equilibrium yield (post-intervention). Thus policy can change system topology by inserting a hard boundary or a new equilibrium (the Fed in a way acts like a hard stop that says yields won't go past X without infinite liquidity coming, thus cutting off the tail risk).

In summary, thinking in terms of manifolds, the system's possible states form a surface where one valley is the low-yield stable region, separated by a ridge from another valley (a disorderly high-yield state). The height of the ridge is lowered by loss of confidence or liquidity (making it easier to slip into the bad valley), and raised by backstops and strong fundamentals (making the good valley encompassing). Monitoring indicators helps gauge where we are on this landscape—approaching a ridge or well within a stable valley. The aim of strategic and policy measures is to keep the system in the desirable equilibrium and to avoid or mitigate those nonlinear jumps across regimes.

## IV. Tooling: Python-Based Monitoring & Sampling

To operationalize the monitoring of the T-bill market and its systemic indicators, we propose a Python-based toolkit. The goal is to use publicly available data (no proprietary feeds or paid subscriptions) to build a pipeline that regularly fetches data, computes indicators, detects anomalies or regime shifts, and provides alerts/visualizations. Such a toolkit can help policymakers or analysts get early warning signals of stress and track the effectiveness of any interventions.

We outline below the components of this monitoring system and provide pseudocode sketches for each:

### A. Data Ingestion

We need to gather time-series data from reliable public sources:

- **Treasury yields:** The Treasury publishes yield curve data (e.g., the Treasury yield curve rates) and the Federal Reserve's FRED database offers daily series for various bill maturities (e.g., 4-week, 3-month T-bill secondary market rates). We can use `pandas_datareader` or the FRED API to pull these. Example series: `DTB4WK` for 4-week bill rate, `DTB3` for 3-month, or `DGS1` for 1-year (though that's a note, not bill, but close).
- **Federal Reserve policy rates:** Important for comparison – e.g., the Fed's Interest on Reserve Balances (IORB) and Overnight Reverse Repo (ON RRP) offering rate. FRED has these (e.g., `IORB` series). The Fed funds effective rate (`DFF`) and OIS rates for short tenors (could use Fed funds futures or proxy via FRED series if available).
- **Repo rates:** A key public series is SOFR (Secured Overnight Financing Rate), available via FRED (`SOFR` series). Also the DTCC publishes GCF repo indices, but SOFR is broad. The Fed's overnight reverse repo facility uptake (usage) is another indicator (FRED: `RRPONTSYD`). Tri-party repo rates (if needed) might be gleaned from New York Fed data releases. For simplicity, we use SOFR and perhaps the repo rate percentiles the NY Fed reports.
- **Auction results:** The Treasury's own site or the fiscaldata API provides auction results data. E.g., bid-to-cover and high yield for each auction. The `fiscaldata.treasury.gov` API's "Treasury Auctions" dataset can be accessed (perhaps via their JSON API) to get recent results. Alternatively, TreasuryDirect

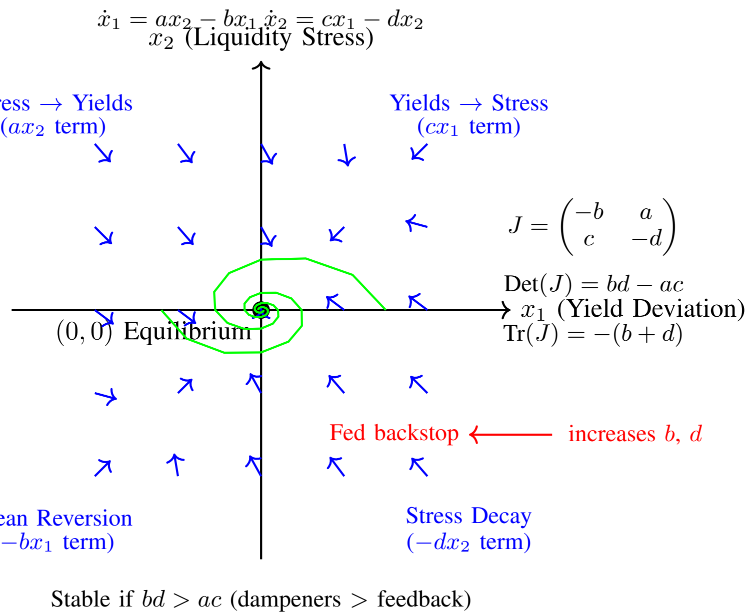


Fig. 5. *Dynamical System with Feedback Loops. The phase plane shows yield deviation ($x_1$) vs liquidity stress ($x_2$). Blue arrows represent the vector field, green curves show stable trajectories spiraling to equilibrium. Stability depends on whether dampening forces (b, d) exceed cross-feedback (a, c). Fed policy intervention can strengthen dampening to maintain stability.*

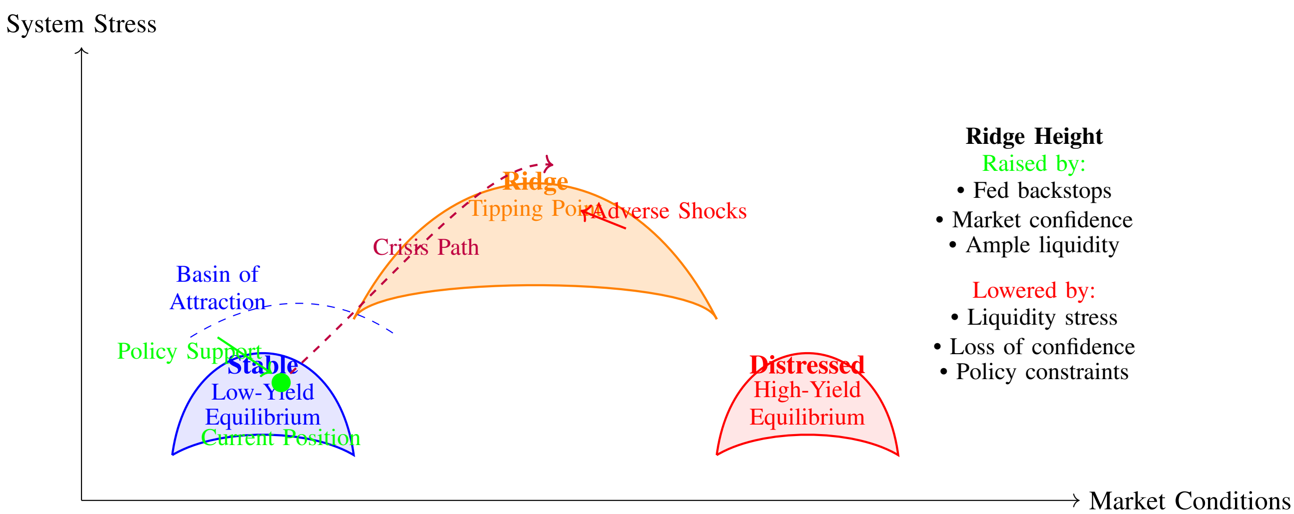


Fig. 6. *Equilibrium Landscape with Valleys and Ridges. The system's state space forms a landscape with a stable low-yield valley (blue) and a distressed high-yield valley (red) separated by a ridge (orange). The height of the ridge determines how easily the system can transition between equilibria. Policy interventions aim to keep the system in the stable valley and raise the ridge height through backstops and confidence measures.*

posts recent results in HTML that can be scraped. Key fields: auction date, security type (e.g., 4-week bill), offering amount, high yield, bid-to-cover, % awarded to different bidder classes. We might fetch the last N auctions for each bill tenor and compute statistics like 3-month trailing average bid-to-cover vs last auction's bid-to-cover.

- **Money Market Fund flows:** The Investment Company Institute (ICI) publishes weekly total assets for government and prime money market funds (often on Wednesday data for week ending). Those could be scraped from their reports or via a CSV if available. Additionally, the OFR has a Money Market Fund Monitor with data (but possibly not easily scriptable without manual download). Alternatively, one could use the SEC Form N-MFP data (monthly, detailed, but heavy to parse). For a simpler approach: high-frequency monitoring might stick to aggregate AUM from a source like ICI or Crane data (if public). Let's assume we can get a weekly time series of government MMF AUM.
- **FX rates:** The broad USD index (Fed's trade-weighted dollar index, e.g., `DTWEXBGS` on FRED) and/or specific

exchange rates like USD/EUR (DEXUSEU on FRED) and USD/JPY can be fetched. We might focus on daily changes in DXY (Dollar Index) versus daily changes in 3-month yield, for example.
- **Treasury market liquidity metrics:** Some may not be directly accessible via API, but proxies like the MOVE index (an index of Treasury implied volatility) is on FRED (MOVEINDEX), which tends to spike when liquidity is poor. Also, primary dealer positions and transactions (Fed's H.8 or weekly dealer holdings report) might provide clues on market-making strain.

We ensure all data is publicly accessible without paywalls. Using Python, one could set up something like:

```
import pandas_datareader.data as web
from datetime import date, timedelta

end = date.today()
start = end - timedelta(days=365*2)  # 2 years
    of data

# Fetch 3-month T-bill yield and SOFR from FRED
tb3m = web.DataReader("DTB3", "fred", start, end
    )   # 3-month bill secondary market rate
sofr = web.DataReader("SOFR", "fred", start, end
    )   # Secured Overnight Financing Rate
ff = web.DataReader("DFF", "fred", start, end)
         # Effective Fed Funds

# Fetch policy rates
iorb = web.DataReader("IORB", "fred", start, end
    )   # Interest on Reserves
onrrp_rate = web.DataReader("RRPONTSYD", "fred",
     start, end)  # RRP award rate

# Fetch broad dollar index
dxy = web.DataReader("DTWEXBGS", "fred", start,
    end)

# (Pseudo-code for auction data, assuming a
    function get_auction_results exists)
auction_df = get_recent_auctions(sec_type="Bills
    ", last_n=20)
# auction_df might have columns: ['issue_date','
    maturity','bid_to_cover','high_yield','
    auction_date',...]
```

*

For auction data, one might need to use `requests` to hit an API endpoint, for example:

```
import requests
import pandas as pd

url = "https://api.fiscaldata.treasury.gov/
    services/api/fiscal_service/v1/debt/auctions
    ?security_type=TBILL"
resp = requests.get(url)
data = resp.json()
df_auctions = pd.DataFrame(data['data'])
```

* (The above endpoint and parameters would need to be adjusted based on actual API documentation, but assume it returns auction records we can filter/sort.)

Data ingestion should be modular, with each data source fetch in a try/except to handle failures, and perhaps caching results (since some data updates daily or weekly, and we don't want to re-download all history each run).

### B. Indicator Computation

Once raw data is in, we compute derived indicators:

- **Spread of T-bill vs policy:** e.g. spr1 $=$ $y_{3m\text{-bill}} -$ Fed funds target or OIS. Since OIS series might not be directly available daily, we could approximate using Fed funds or simply use Fed's Interest on Reverse Repo (which sets a floor) or IORB. For simplicity, say spr1 $=$ $y_{3m} -$ IORB. We would then look at spr1's history (perhaps normalized by its standard deviation).
- **Auction trend metrics:** For each bill tenor, compare latest bid-to-cover with its 1-year median. Compute percentile or z-score. Also track any trend (like is it consistently falling last 3 auctions?). Similarly, compute the tail: difference between auction high yield and when-issued yield just before auction. If we have that data, tail $> 0$ is a weaker auction. We might flag if tail is above some threshold (e.g., more than 2 standard deviations of past tails).
- **Repo divergence:** Compare SOFR with IORB (normally SOFR should be a few bps above ON RRP floor but within Fed's target range). If SOFR spikes $>$ target range or relative to bills, flag it. Also monitor SOFR-FF spread. And if available, volume of Fed's repo operations or RRP uptake (though RRP uptake is more a daily liquidity absorption figure – large uptake usually when banks are full, not necessarily stress).
- **Fails:** If we can get data on Treasury settlement fails (New York Fed publishes weekly aggregate fails in their statistical release), we'd include that. If fails $>$ some millions threshold and rising, it's a stress sign.
- **MMF flows:** Compute weekly (or daily if available) change in government MMF AUM. If huge inflow (say top 5% percentile of changes), that indicates risk aversion rising (money seeking safety). If huge outflow, perhaps money leaving due to yield chasing elsewhere or concerns. Also track MMF usage of the Fed's ON RRP facility (published daily). If ON RRP usage is trending up to record highs, it means MMFs prefer parking cash at Fed rather than bills – possibly bills yields too low or scarce supply. That can indicate a collateral shortage (or Fed's rate being attractive floor). A decline in RRP usage in favor of bills can indicate tightening conditions (bills yielding more so MMFs shift to market).
- **FX-yield correlation:** We could compute a rolling correlation between, say, daily changes in 2-year Treasury yield and daily changes in DXY (dollar index). In normal times, this correlation might be positive (higher yields $\rightarrow$ stronger dollar). If it turns negative strongly (with yields and USD moving opposite in a notable way), that might be an indicator foreigners are acting. This might be more of an ad-hoc check rather than a formal indicator. Alternatively, monitor the DXY level and 3m yield spread vs G7 average yield – unusual deviations might signal something.
- **Composite Stress Index:** Create a weighted index com-

bining several normalized indicators, to summarize the overall stress level in the T-bill/liquidity system. For example, we can z-score each of: bill-OIS spread, SOFR-IORB spread, bid-to-cover deviation, MMF flow, FX basis (if we have it), etc., then take a weighted sum. Weights could be based on their typical volatility or historical impact. The composite index might be calibrated to have mean 0, std 1 in normal times, and see how many standard deviations it moves. This can give a single time-series to track, akin to an "UST liquidity stress index".

We should include logic for **regime shift detection**: One approach is to use rolling statistical tests (e.g., a sudden jump in variance or mean of certain indicators). For example, implement a CUSUM test or a change-point detection on the composite index. If the composite breaks beyond, say, the 95th percentile of its calm distribution, that signals entering a stress regime. Or if correlations that used to hold (like yields vs USD) invert consistently over several days, flag a possible regime change.

Additionally, **divergence detection**: For instance, if short-term yields are rising but long yields are falling (a sign of maybe a policy credibility issue or flight to safety in long end while funding stress in short end), that divergence can be automatically flagged. Or the mentioned yields vs USD opposite movement. We could code a check:

```
if tb3m.pct_change() > 0.005 and dxy.pct_change
    () < -0.002:
    flag_foreign_outflow = True
```

* (for example, 0.5% rise in bill yield with 0.2% drop in dollar index in a day might be unusual).

The Python implementation can simply compute these conditions each day and set boolean flags or scores.

Pseudocode example for computing and flagging:

```
import numpy as np

# Compute 3m bill - IORB spread
spr1 = tb3m['DTB3'] - iorb['IORB']
spr1_latest = spr1.iloc[-1]
spr1_z = (spr1_latest - spr1.mean())/spr1.std()

# Auction bid-to-cover z-score for 4-week bills
bc_4w_latest = auction_df[auction_df['
    security_term']=='4-week']['bid_to_cover'].
    iloc[-1]
bc_4w_mean = auction_df[auction_df['
    security_term']=='4-week']['bid_to_cover'].
    mean()
bc_4w_std  = auction_df[auction_df['
    security_term']=='4-week']['bid_to_cover'].
    std()
bc_4w_z = (bc_4w_latest - bc_4w_mean)/bc_4w_std

# Repo vs IORB
sofr_latest = sofr.iloc[-1]
iorb_rate = iorb.iloc[-1]
repo_spread = sofr_latest - iorb_rate

# Example flags
stress_flags = {}
stress_flags['bill_spread_high'] = spr1_z > 2  #
     beyond 2 std dev
stress_flags['weak_auction_4w'] = bc_4w_z < -2
    # bid-cover 2 std below mean (weak demand)
stress_flags['repo_spike'] = repo_spread > 0.25
     # SOFR 25bp above IORB, abnormal

# FX divergence
dxy_chg = dxy.iloc[-1] - dxy.iloc[-2]
yld_chg = tb3m.iloc[-1] - tb3m.iloc[-2]
stress_flags['yields_up_usd_down'] = (yld_chg >
    0.05) and (dxy_chg < -0.3)

# Composite index
# e.g., weight the z-scores of key metrics
comp = 0.25*spr1_z + 0.25*bc_4w_z*(-1) + 0.25*((
    sofr_latest-iorb_rate) / 0.05) + 0.25*(
    mmf_flow_z)
comp_index = comp  # simple weighted sum for
    interpretation
```

*

(The specific thresholds and weights would be tuned based on historical data).

*C. Visualization and Alerts*

The toolkit should produce outputs that analysts can quickly understand:

- **Time series plots:** e.g., a chart of the 3-month bill vs RRP rate, highlighting when it diverges; a plot of bid-to-cover over time with a band for normal range; a composite stress index chart with shading for past stress episodes (so current level can be compared visually).
- **Tabular summary of latest readings:** e.g., "Bid-to-cover 4w: 2.8 (median 3.2) – low" or "SOFR – IORB spread: +5 bps (normal ˜0-3 bps)".
- **Possibly a heatmap of indicators** (green = normal, yellow = moderate, red = extreme) for the current day.
- **If automated alerting is needed,** the code can trigger conditions. For instance, if more than 3 of the predefined flags are true, or if composite index > certain value, it could print an alert or send an email (depending on implementation environment).

For example:

```
if comp_index > 3 or sum(val for val in
    stress_flags.values() if val) >= 3:
    print("ALERT: Multiple stress indicators
        triggered. Composite index = {:.2f}".
        format(comp_index))
    if stress_flags['weak_auction_4w']:
        print(" - Recent 4-week bill auction was
            very weak.")
    if stress_flags['repo_spike']:
        print(" - Repo rate significantly above
            normal, indicating funding strain.")
    # etc.
```

*

This text could then be included in a daily report.

Assumptions and limitations: All data is public, which means some granularity is lost (e.g., we don't have intraday monitoring of bids or the full depth data a central bank might have). We rely on end-of-day or weekly data mostly, so very rapid flash events (like an intraday repo spike) might only be seen after the fact (though SOFR capturing overnight average at least). Also, our approach assumes historical norms are a good benchmark (z-scores etc.), which might need recalibration if the regime changes (one might need to use expanding window or a fixed pre-crisis baseline to identify when things are unusual). The toolkit also doesn't explicitly model causality – it flags correlations and deviations, but human analysis is needed to interpret if, say, a repo spike is quarter-end technical vs true stress.

Nonetheless, by automating data collection and having codified thresholds, the Python toolkit ensures we don't miss warning signs due to human complacency, and it allows consistent tracking over time. The modular design (functions for fetching each dataset, computing each indicator, etc.) makes it maintainable – e.g., if a new indicator becomes relevant (like a new Fed facility usage), we can add it easily.

### D. Example: Weekly Monitoring Routine

To illustrate, a weekly run (say every Friday) could output something like:

- "Composite Stress Index: 1.7 (on scale where >3 is high stress). This is up from 0.5 last week, highest since March 2024."
- Bullet points per indicator:
  - "T-bill spread to RRP: +12 bps (widened from +5 bps last week, 90th percentile level) – indicating bills selling off relative to Fed's floor."
  - "4-week bill auction tail: 4 bps (significant tail, recent auctions showed reduced demand, bid-to-cover 2.5 vs 12-month avg 3.1)."
  - "Repo rate (SOFR) averaged 5.10%, 5 bps over IORB – slightly elevated, likely quarter-end effects."
  - "MMF flows: +$30B into government funds this week – large inflow, suggests risk aversion."
  - "Dollar index down 1% even as 3m yield rose from 4.5% to 4.7% – possible foreign selling signal."
- And maybe a short interpretation: "Overall signs of moderate stress emerging: bill yields have moved up notably above policy rates, and auction metrics deteriorated. Situational awareness warranted, though not yet at crisis levels. If these trends continue or intensify next week, risk of self-reinforcing liquidity strain increases."

This leads naturally to the next section: how an AI like ChatGPT could take such data and produce daily commentary under strict rules.

## V. Meta-Layer: Instructing ChatGPT as a Daily Sentinel

Finally, we describe how one could harness a language model (LLM) like ChatGPT to act as a "sentinel" that reviews the latest data each day and produces an analysis for human oversight. The key is to give the AI a well-defined role, scope, and set of rules so that its output is reliable, nuanced, and free of false alarms. We propose a structured **system prompt** that can be reused, which includes instructions on scanning, interpretation logic, output format, and guardrails.

Below is a *draft system prompt* one could provide to ChatGPT (or similar) each day before feeding it the updated indicators:

System Role and Context: You are an analyst LLM tasked with monitoring the U.S. Treasury bill and related funding markets on a daily basis. You have access to the latest public data on key indicators (T-bill yields, auction results, repo rates, money market fund flows, FX rates, etc.). Your goal is to identify any signs of liquidity stress, shifts in market equilibrium, or unusual activity that could indicate changes in systemic stability.

You are not a trading advisor; you are a risk sentinel for policymakers. You should remain objective, focusing on facts and measured analysis. The time frame is daily (with weekly synthesis on Fridays), so focus on short-term changes in context of longer trends.

Daily Scanning Tasks:

1) Data Ingestion: Each morning, read in the fresh values of indicators including:
   - Short-term Treasury yields (e.g., 1m, 3m T-bill rates) vs policy rates (Fed RRP/IOR).
   - Recent Treasury auction metrics (bid-to-cover ratios, tail amounts, etc.).
   - Repo market rates (SOFR or relevant overnight rates) and any reported anomalies (e.g., usage of Fed facilities).
   - Money market fund flows or RRP usage changes.
   - FX market moves (especially USD strength-/weakness relative to interest rate moves).
   - Any news of note (e.g., Fed statements, debt ceiling developments, etc. if provided).
2) Historical Baseline Comparison: Compare today's values to recent history (e.g., yesterday, last week, normal range). Identify which changes are noteworthy (e.g., an indicator moving beyond typical variation). Use z-scores or percentile where applicable to gauge magnitude.
3) Flag Anomalies: Determine if any cluster of indicators points to emerging stress:
   - e.g., Bill-OIS spread significantly widened and repo rate spiked together (indicative of funding strain),
   - or multiple weak auctions in a row,
   - or an unusual combination like rising yields with falling dollar (foreign selling).

   Flag these in your analysis.

Interpretation Rules:

- Use a corroborative approach: avoid drawing conclusions from a single indicator in isolation. Require either a very extreme single move or confirmation from other metrics before making a strong assessment.
- Distinguish between likely transient "noise" (e.g., quarter-end effects causing a one-day repo jump) and structural signals (e.g., a persistent upward trend in bill yields over several days out of line

with policy rate).

- Always consider plausible explanations for anomalies (technical factors, policy statements, etc.) and note uncertainties.
- Do not make deterministic predictions. Instead, assess risks and scenarios ("if X continues, it could indicate Y").
- Avoid emotive or alarmist language. Use measured terms (e.g., "elevated", "notable", "needs monitoring") rather than "crisis" or "crashing", unless truly warranted by multiple severe signals.
- Emphasize mechanism: if you note a pattern (say, foreign selling indications), mention the mechanism (e.g., "this pattern could reflect foreign reserve managers reducing exposure, as seen by...") to justify the interpretation.

Output Format Requirements:

- Daily Summary (Bullet Points): Provide 3–6 bullet points summing up the key observations for that day. Each bullet should be concise (1–2 sentences) and factual. Example:
  - "3-month T-bill yield rose to 4.75%, widening its spread over the Fed's RRP rate to 20 bp (highest in 6 months), indicating reduced demand for bills."
  - "Today's 4-week bill auction saw a bid-to-cover of 2.4, below recent average of ˜3.0, and tailed by 5 bp — a weak result that suggests investor caution at the short end."
  - "SOFR jumped to 5.1% (from 5.0%), briefly exceeding the Fed's IOR, likely due to quarter-end funding pressures; usage of the Fed's repo facility also ticked up."
  - "Government money market funds had an inflow of +$10B, consistent with a move into cash, while the USD index fell 0.5% even as yields rose – a combination that may imply some foreign selling of Treasuries."
- Weekly Synthesis (Fridays): In addition to daily points, provide a brief paragraph (3–5 sentences) on weekly trends and overall risk assessment. Summarize whether conditions improved, deteriorated, or stayed stable over the week, and highlight any emerging theme (e.g., "liquidity is tightening", or "market absorbed increased supply well"). Include an explicit confidence level (e.g., "Confidence: Moderate – the signals are somewhat mixed, so interpretation is cautious.").
- Confidence and Uncertainty: State confidence qualitatively (low/medium/high) regarding any assessments. If uncertain, clearly say so (e.g., "It's unclear if this is a one-off or start of a trend."). If a potential issue is identified, articulate what would confirm or allay it (e.g., "We will watch next week's auctions for confirmation of weak demand.").

Guardrails:

- No Blind Guessing: If data is missing or inconclusive, do not speculate wildly. Either state that data was not available or that results are inconclusive.
- No Predictions Without Rationale: Avoid any forward-looking statements unless grounded in a clear mechanism. For example, saying "If bill-OIS spreads continue widening, the Fed may intervene" is acceptable *because* you're linking to a mechanism; saying "We think a crash is coming" without evidence is not.
- Fact/Inference/Speculation Distinction: Clearly separate what is directly observed (facts: "X happened"), what is an interpretation (inference: "X could imply Y"), and what is a cautious forward view if needed ("If Y persists, then Z might occur."). Use language like "suggests", "could indicate", "possibly" for inferences; use "will" or "is" only for established facts.
- Stay on Topic: Focus only on aspects relevant to T-bill market stability and liquidity. Do not deviate into unrelated macro news or political commentary except if directly impactful (e.g., a debt ceiling standoff news is relevant, but an unrelated stock market move is not to be analyzed here).
- Maintain Professional Tone: The audience is economists and policy folks, so use clear, formal language, avoid exaggeration, and ensure any technical terms are used correctly.

End of prompt.

This system prompt ensures the AI operates within strict guidelines: it looks at clusters of indicators, avoids single-indicator traps, quantifies changes against baselines, and is conservative in its conclusions. It essentially makes the AI follow the same analytical discipline we have outlined in this report.

Using such a prompt daily, one could have ChatGPT ingest the latest numbers (by either feeding those as part of user prompt or connecting to data), and then the model would output something like:

(Sample AI output following the above rules):

- "Daily Summary (Thu Sep 24, 2026):
  - The 3m T-bill yield increased to 5.10% ( +15 bps day-on-day), widening its spread over the Fed's overnight RRP rate to ˜30 bps – an atypically large gap suggesting softer demand for bills.
  - Today's 8-week bill auction was soft: bid-to-cover ratio of 2.6 vs a 3.1 average, and the stop-out yield came 4 bps above the when-issued yield, indicating investors required a premium.
  - SOFR rose to 5.15% (up 10 bps) and temporarily exceeded the Fed's interest on reserves, pointing to quarter-end funding strains; Fed's repo facility saw $5B usage (first usage in months).

- Government MMFs saw approx. \$9B inflow (week-to-date), consistent with a cautious shift into cash. Notably, the US dollar index fell 0.3% today despite higher U.S. short-term rates, a pattern that could reflect foreign accounts trimming Treasury holdings.
- Weekly Trend & Risk (Week of Sep 20–24): Short-end funding conditions have tightened somewhat this week. T-bill yields steadily rose and auction demand showed signs of fatigue, while dollar funding rates firmed up. These moves partly reflect typical quarter-end dynamics, but the magnitude is larger than usual (confidence in this assessment: medium). If next week's auctions and funding rates normalize, it would suggest this was transient; otherwise, it may signal emerging liquidity tightness that warrants close monitoring, especially in the context of ongoing Fed balance sheet reduction."

Such an output addresses the needs: it's factual, multi-indicator, not alarmist but also not complacent, and explicitly notes confidence and uncertainty.

This daily AI-driven briefing could be a powerful tool for decision-makers, acting as an early warning system that synthesizes complex data into a coherent snapshot with the reasoning attached. The guardrails ensure it doesn't cry wolf inappropriately or miss subtle but important shifts by focusing on clusters and context.

## Conclusion

In this report, we treated the U.S. T-bill market as a critical control surface of the global dollar system – one that can be influenced by various actors and whose state reveals a great deal about underlying liquidity and confidence. We mapped the system's structure, examined strategic behaviors of key players, defined the latent variables and observables of the system, and proposed both quantitative models and practical tooling to monitor stability. Our analysis underscores clear separations between factual observables (e.g., yields, flows, spreads), the inferred latent state (liquidity stress, confidence levels), and strategic implications (the potential for adversaries to exploit or allies to stabilize). We also delineated assumptions (such as rationality of actors within their constraints) and noted where these could break down (failure cases like policy impotence in high inflation, or misjudgment of an adversary's resolve).

Finally, we translated these insights into a blueprint for a monitoring system – both code-based and AI-based – to make this actionable. The Python toolkit gathers and quantifies the signals, and the ChatGPT prompt ensures a disciplined interpretive layer for daily use. By implementing these, economists and engineers can have a daily sentinel that flags brewing issues in T-bill markets before they snowball, helping maintain the integrity of this vital financial nerve center.

## Acknowledgments

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## About This Document

This document represents a technical exploration created through a collaborative process between human and AI. The production process followed these steps:

1) **Discovery**: [Describe how the topic was discovered or what motivated the exploration]
2) **Initial Exploration**: [Describe the initial research and dialogue process with AI systems]
3) **Synthesis**: [Describe how the dialogue results were organized into a structured document]
4) **Implementation**: [If applicable, describe any code or practical implementations developed]
5) **Visualization**: [If applicable, describe diagram and figure creation process]
6) **Verification**: All references were scrutinized for authenticity. URLs were tested for accessibility, author names were verified, and content relevance was checked against citations. This verification process is documented in the following section.

[Add a concluding statement about the document's purpose and intended audience]

## Note on References and Verification

*This document contains AI-generated content. All references have been subject to rigorous verification to ensure academic integrity.*

**Verification Process:**

- All URLs were tested for accessibility using automated tools
- Author names were verified against real publications
- DOIs were confirmed where available
- Publication venues (journals, conferences) were validated
- Content relevance was checked against citations

**Verification Status in References:** Each reference includes a `note` field indicating its verification status:

- "Verified: URL accessible" – URL was tested and works
- "Verified: DOI accessible" – DOI was confirmed
- "Standard reference" – Well-known textbook or established work
- "Requires verification" – Needs manual review

**Important Notice:** Due to the AI-assisted nature of this document's creation, readers should independently verify any references used for critical applications.

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